

SMITHIAN FOLD THEORY - SOCIAL AND COLLECTIVE SCIENCES PAPER 001

From One Relation to Society

An Exact, Zero-Parameter and Machine-Closed Foundational Reconstruction of
Social and Collective Sciences from Smithian Fold Theory

Ernos Labs

Open Source Science Platform and Knowledge Tree

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Version 1.0.0 - current-evidence closed, extension-open foundation

72 required laws - 18,432 exact candidates

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Abstract

Social science fails when a convenient aggregate is allowed to erase its people, when an institutional label stands in for an observed process, when prestige becomes evidence, when dissent becomes error by declaration, or when an ethical preference is hidden inside an empirical claim. This paper reconstructs the foundational Social and Collective Sciences branch of Smithian Fold Theory from the One through admitted mathematical, informational, computational, biological, consciousness, medical, Earth and astronomical dependencies. The result comprises twelve families and seventy-two laws governing agents, groups, population, observation, communication, coordination, conflict, institutions, trust, knowledge, exchange, distribution, power, networks, culture, multi-level causation, history, intervention, ethics and disciplinary handoff. The proof layer adds no axiom, free parameter, fit, conventional numerical zero, negative proof scalar, irrational or imaginary proof number, completed infinity or ungenerated continuum. Every claim is a literal 256-form census across eight binary axes. External records are opened only after the complete 18,432-candidate branch seal. Each result is then independently reconstructed and compared against source-custodied evidence while adverse, unavailable and unresolved records remain visible. The paper gives the full chain for every claim and establishes an extension-open empirical constitution for collective inquiry.

Authorship, access and the public scientific mission

Maria Smith produced this work outside credentialed institutional access. That fact is not offered as personal exceptionalism. It is an indictment of all the minds and contributions society loses when financial gatekeeping, credentials, paywalls, prestige, institutional permission and access to funded infrastructure are dressed as rigor. The appropriate response is not to ask whether an excluded author was somehow uniquely permitted to think. It is to ask how many minds, observations and corrections were never heard because the gate was mistaken for the method.

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Open criticism requires no credential. Scientific admission requires the common evidence protocol. Contact: **Maria.Smith.Sftoe@gmail.com**. Public submissions and review: <https://discord.gg/ucwGryVxGr>. GitHub: <https://github.com/MettaMazza>.

Mathematical constitution

Every law traces to **There Is No Nothing** and to already admitted Fold dependencies. No additional axiom is introduced. No free or fitted parameter can select a survivor. Proof-state arithmetic is exact and finite. Conventional numerical zero is not a proof carrier; the glyph $\emptyset$ may be retained as a typed marker for absence, non-response or a source field, but it does not import ordinary zero semantics. Negative, irrational and imaginary quantities may appear as held source labels where an external dataset uses them, but never as proof scalars. Completed infinities and ungenerated continua are excluded.

For each claim the generator builds eight binary dimensions: carrier, boundary, relation, record, evidence, provenance, generality and extension. Their literal Cartesian product contains $2^8 = 256$ candidates. One choice on each dimension alone preserves its required distinction. The program decides all 256 candidates; one survives and 255 are eliminated. Across 72 claims this is exactly 18,432 decided forms. A separate implementation rebuilds the product and survivor from the declared axes. Positive-finite induction proves that adding one lawful agent, relation, context, period or record preserves all earlier identities, adverse rows and boundaries while appending its trace.

This closure is exact but bounded. It establishes the foundational forms necessary to make Social claims reconstructible and falsifiable. It does not prohibit new societies, datasets, institutions, forms of exclusion, historical sources or lawful discoveries. The

branch is therefore current-evidence closed and extension-open, never permanently locked.

Empirical constitution and blind order

The complete inventory, candidate grammar, predicted structural labels and falsification conditions were frozen before any external source identity or outcome was opened. Primary custodians were selected afterward for their custody of relevant observation, survey, demographic, conflict, institutional, health, education, publication and historical records-not because their institutional status could prove a law. A capability-closed prediction process cannot read a file, network, clock, environment variable, process or target. Only after its derivation seal exists may the external evaluator open the registered target.

Every comparison retains successful, adverse, absent, censored, missing, disputed, unresolved, access-denied and transport-failed rows. An inaccessible archive is not evidence of no event. A source's classification is not automatically truth. A survey response is not a private mental state. An institutional index is not the institution itself. A model, estimate or forecast remains a model, estimate or forecast. These distinctions are part of the empirical result, not administrative bookkeeping.

The immutable engine seal is

sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a. The

verification-authority seal is

sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8. A mismatch halts.

Those identities secure a shared admission route; they never replace the readable derivations.

Dependency and ownership boundary

Mathematics owns exact graph, order, probability and optimization structures. Computation owns abstract communication, causality, consensus and distributed knowledge. Biology owns organism-level processes. Consciousness owns first-person experience and qualia. Medicine owns clinical state. Earth Science owns physical hazards and environments. Social and Collective Sciences owns measured relations, group membership, institutions, distribution, collective action, historical context, public access and social consequence. Engineering Translation may implement and test applications but cannot select or rewrite the laws. This ownership map prevents an abstract edge from becoming a social tie, a simulated agent from becoming a human observation, or a product metric from becoming scientific admission.

Registered primary-source surface

- **UNSD-DEMOGRAPHIC-001 - United Nations Statistics Division.** Demographic and Social Statistics. Registered families: agents_groups_identity, exchange_allocation_distribution, power_authority_exclusion. Locator: https://unstats.un.org/unsd/demographic-social/products/dyb/dyb_2024/. Its authority here is source custody and documented provenance, not prestige or consensus.
- **WORLD-BANK-WDI-001 - World Bank Data Group.** World Development Indicators. Registered families: exchange_allocation_distribution, power_authority_exclusion, multi_level_causation. Locator: <https://datacatalog.worldbank.org/search/dataset/0037712/world-development-indicators>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **ILOSTAT-001 - International Labour Organization Department of Statistics.** ILOSTAT data. Registered families: exchange_allocation_distribution, power_authority_exclusion, agents_groups_identity. Locator: <https://ilostat.ilo.org/data/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **UNESCO-UIS-001 - UNESCO Institute for Statistics.** UIS Data Browser. Registered families: culture_language_transmission, exchange_allocation_distribution, power_authority_exclusion. Locator: <https://databrowser.uis.unesco.org/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **UCDP-DOWNLOADS-001 - Uppsala Conflict Data Program.** UCDP Dataset Download Center. Registered families: coordination_conflict, history_intervention_ethics, power_authority_exclusion. Locator: <https://ucdp.uu.se/downloads/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **VDEM-DATA-001 - Varieties of Democracy Institute.** V-Dem Dataset Archive. Registered families: norm_institution_enforcement, power_authority_exclusion, history_intervention_ethics. Locator: <https://v-dem.net/data/dataset-archive/>. Its authority here is source custody and documented provenance, not prestige or consensus.

- **IPUMS-INTERNATIONAL-001 - IPUMS / University of Minnesota.** IPUMS International. Registered families: agents_groups_identity, exchange_allocation_distribution, culture_language_transmission, multi_level_causation. Locator: <https://www.ipums.org/projects/ipums-international>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **GSS-NORC-001 - NORC at the University of Chicago.** General Social Survey. Registered families: trust_reputation_knowledge, culture_language_transmission, observation_action_communication. Locator: <https://gss.norc.org/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **OECD-DATA-EXPLORER-001 - Organisation for Economic Co-operation and Development.** OECD Data Explorer. Registered families: exchange_allocation_distribution, norm_institution_enforcement, multi_level_causation. Locator: <https://data-explorer.oecd.org/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **UNHCR-REFUGEE-DATA-001 - United Nations High Commissioner for Refugees.** Refugee Data Finder. Registered families: agents_groups_identity, power_authority_exclusion, history_intervention_ethics. Locator: <https://www.unhcr.org/refugee-statistics/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **UNODC-DATA-001 - United Nations Office on Drugs and Crime.** UNODC Data Portal. Registered families: norm_institution_enforcement, coordination_conflict, history_intervention_ethics. Locator: <https://dataunodc.un.org/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **WHO-GHO-001 - World Health Organization.** Global Health Observatory. Registered families: multi_level_causation, ownership_handoffs, exchange_allocation_distribution. Locator: <https://www.who.int/data/gho>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **ICPSR-DATA-001 - Inter-university Consortium for Political and Social Research.** ICPSR Data Archive. Registered families: observation_action_communication, trust_reputation_knowledge, history_intervention_ethics. Locator: <https://www.icpsr.umich.edu/web/pages/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **CROSSREF-RECORDS-001 - Crossref.** Crossref metadata and records. Registered families: trust_reputation_knowledge, culture_language_transmission, ownership_handoffs. Locator: <https://www.crossref.org/documentation/retrieve-metadata/rest-api/>. Its authority here is source custody and documented provenance, not prestige or consensus.

Family results

Agents, relations, groups and collective identity

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Social agent** (SFT-SOCIAL-AGENT-001): A social agent is a bounded acting and observed carrier whose identity, capability, context and alternatives remain explicit.
- **Social relation** (SFT-SOCIAL-RELATION-001): A social relation is a typed, directed or reciprocal interaction between identified agents in a declared context.
- **Group identity** (SFT-SOCIAL-GROUP-001): A group retains membership, boundary, purpose, entry, exit and internal disagreement rather than becoming an indivisible person.
- **Collective state** (SFT-SOCIAL-COLLECTIVE-001): A collective state composes agent states and relations while retaining internal distribution and minority states.
- **Organization identity** (SFT-SOCIAL-ORGANIZATION-001): An organization is a versioned arrangement of members, roles, rules, resources and processes.
- **Social population** (SFT-SOCIAL-POPULATION-001): A social population is a sampling-frame-bound agent set with selection, weighting, censoring and missingness retained.

Observation, action and communication

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Social observation** (SFT - SOCIAL - OBSERVATION - 001): Social observation retains observer, method, consent, setting, reactivity and unavailable records.
- **Social action** (SFT - SOCIAL - ACTION - 001): Action attribution retains capability, alternatives, constraints, intent evidence, act and consequence.
- **Communication event** (SFT - SOCIAL - MESSAGE - 001): A communication event retains sender, content, channel, recipient opportunity, reception and loss.
- **Shared social record** (SFT - SOCIAL - SHARED - RECORD - 001): A shared record is a versioned artifact with authorship, access, custody, changes and disputes retained.
- **Common-knowledge boundary** (SFT - SOCIAL - COMMON - KNOWLEDGE - 001): Common knowledge claims require a declared finite mutual-observation depth and complete receipt chain.
- **Social communication noise** (SFT - SOCIAL - COMMUNICATION - NOISE - 001): Communication noise retains technical loss, ambiguity, reinterpretation and audience differences.

Coordination, cooperation, competition and conflict

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Coordination** (SFT - SOCIAL - COORDINATION - 001): Coordination is compatible action around a declared target with dependencies, timing and failures retained.
- **Cooperation** (SFT - SOCIAL - COOPERATION - 001): Cooperation retains contributions, costs, benefits, alternatives and defection for identified participants.
- **Competition** (SFT - SOCIAL - COMPETITION - 001): Competition is a rule- and resource-bounded rival claim with opportunities and exclusion retained.
- **Conflict** (SFT - SOCIAL - CONFLICT - 001): Conflict retains incompatible claims, capabilities, actions, harms, alternatives and time.
- **Bargaining** (SFT - SOCIAL - BARGAINING - 001): Bargaining retains offers, outside options, information, power, time, breakdown and agreement.
- **Collective action** (SFT - SOCIAL - COLLECTIVE - ACTION - 001): Collective action retains participation, abstention, contribution, threshold, benefit, failure and context.

Norms, rules, institutions and enforcement

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Social norm** (SFT - SOCIAL - NORM - 001): A norm is a context-bound expectation evidenced by behavior and responses to conformity and violation.
- **Social rule** (SFT - SOCIAL - RULE - 001): A rule retains issuer, text, version, scope, subjects, exceptions, application and contestation.
- **Institution** (SFT - SOCIAL - INSTITUTION - 001): An institution is a historically repeated governed process, not merely a building, name or consensus.
- **Enforcement** (SFT - SOCIAL - ENFORCEMENT - 001): Enforcement retains monitoring, selectivity, procedure, response, appeal and non-enforcement.
- **Compliance and resistance** (SFT - SOCIAL - COMPLIANCE - 001): Compliance and resistance remain distinct from ignorance, inability, concealment and coincidence.
- **Institutional change** (SFT - SOCIAL - INSTITUTIONAL - CHANGE - 001): Institutional change is a source-dated transition retaining continuity, conflict and alternative accounts.

Trust, reputation and collective knowledge

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Trust relation** (SFT - SOCIAL - TRUST - 001): Trust is domain-specific reliance under vulnerability with history, action, outcome and breach retained.
- **Reputation** (SFT - SOCIAL - REPUTATION - 001): Reputation is a network of attributed reports whose sources, incentives, conflicts and audiences remain visible.
- **Testimony and credibility** (SFT - SOCIAL - TESTIMONY - 001): Testimony retains speaker access, claim, support, conflicts, audience and reception without credential substitution.
- **Collective knowledge** (SFT - SOCIAL - COLLECTIVE - KNOWLEDGE - 001): Collective knowledge is distributed, recorded and validated content with contribution, access and disagreement retained.
- **Expertise evidence boundary** (SFT - SOCIAL - EXPERTISE - 001): Expertise is domain- and task-bound competence evidenced by methods and performance, not a universal authority token.
- **False and unsupported social information** (SFT - SOCIAL - MISINFORMATION - 001): False, unsupported, disputed and corrected claims remain distinct and source-dated.

Exchange, allocation and distribution

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Exchange** (SFT - SOCIAL - EXCHANGE - 001): Exchange retains parties, items, rights, terms, timing, costs and failed transfers.
- **Social valuation** (SFT - SOCIAL - VALUE - 001): Valuation is agent-, purpose-, option- and context-bound and cannot become an intrinsic universal scalar by declaration.
- **Resource allocation** (SFT - SOCIAL - ALLOCATION - 001): Allocation retains total resources, claimants, rule, priority, outcome, unmet claims and losses.
- **Collective production** (SFT - SOCIAL - PRODUCTION - 001): Production retains labour, material, knowledge, ownership, transformation, output and loss.
- **Distribution and inequality** (SFT - SOCIAL - DISTRIBUTION - 001): Distribution retains population, total stock, shares, access, time and missing members.
- **Social mobility** (SFT - SOCIAL - MOBILITY - 001): Mobility is a cohort- and measure-bound transition retaining paths, attrition and opportunity structure.

Power, authority, dependency and exclusion

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Power relation** (SFT - SOCIAL - POWER - 001): Power is a directional capacity to alter another's available actions, retaining resources, dependency and resistance.
- **Authority** (SFT - SOCIAL - AUTHORITY - 001): Authority is scoped decision capacity with source, procedure, recognition and contestation retained.
- **Social dependency** (SFT - SOCIAL - DEPENDENCY - 001): Dependency retains the needed resource, controller, alternatives, costs and failures.
- **Exclusion** (SFT - SOCIAL - EXCLUSION - 001): Exclusion retains opportunity, eligibility, gate, decision, appeal and unobserved discouraged cases.
- **Stratification** (SFT - SOCIAL - STRATIFICATION - 001): Stratification is persistent ordered access across a declared population and resource set with mobility retained.
- **Scientific and institutional gatekeeping** (SFT - SOCIAL - GATEKEEPING - 001): Gatekeeping evidence retains criteria, costs, credentials, decisions, appeals and contributions made invisible by exclusion.

Networks, diffusion and collective transition

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Social network** (SFT - SOCIAL - SOCIAL - NETWORK - 001): A social network retains node identity, link meaning, direction, time, sampling and missing links.
- **Social diffusion** (SFT - SOCIAL - DIFFUSION - 001): Diffusion retains source, path, exposure opportunity, adoption, nonadoption and external inputs.
- **Collective synchronization** (SFT - SOCIAL - SYNCHRONIZATION - 001): Synchronization is recurrent phase relation with coupling, lag, breaks and common-cause controls retained.
- **Collective cascade** (SFT - SOCIAL - CASCADE - 001): A cascade retains ordered local transitions, paths, nontransitions and external causes.
- **Collective transition** (SFT - SOCIAL - COLLECTIVE - TRANSITION - 001): Collective transition is a population-distribution change with drivers, controls, reversals and historical boundary retained.
- **Socio-economic cycle** (SFT - SOCIAL - SOCIAL - CYCLE - 001): A social cycle requires recurrence under a declared state definition and retains shocks, broken cycles and measurement change.

Culture, language and transmission

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Culture** (SFT - SOCIAL - CULTURE - 001): Culture is historically transmitted and varied practice and meaning, not an immutable property of a labelled group.
- **Social symbol and meaning** (SFT - SOCIAL - SYMBOL - 001): A symbol's social meaning is user-, context- and history-bound with alternatives retained.
- **Language community** (SFT - SOCIAL - LANGUAGE - COMMUNITY - 001): A language community retains speakers, varieties, contexts, power, change and corpus limits.
- **Education and knowledge transmission** (SFT - SOCIAL - EDUCATION - 001): Education evidence retains access, instruction, participation, assessment, attrition and context.
- **Media system** (SFT - SOCIAL - MEDIA - 001): A media system retains ownership, production, selection, distribution, reach, response and missing exposure.
- **Collective memory** (SFT - SOCIAL - COLLECTIVE - MEMORY - 001): Collective memory is a contested transmission of records, narratives and practices with omissions retained.

Individual-collective composition and causation

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Individual-to-collective composition** (SFT - SOCIAL - MICRO - MACRO - 001): Individual states compose a collective only through an explicit relation and aggregation rule retaining lost distinctions.
- **Collective-to-individual constraint** (SFT - SOCIAL - MACRO - MICRO - 001): Collective structures affect individuals through observable rules, resources, exposure and expectations.
- **Collective emergence** (SFT - SOCIAL - EMERGENCE - 001): An emergent collective property is composition-dependent and not identical to any one component, with tests and scale retained.
- **Human and artificial multi-agent collective** (SFT - SOCIAL - MULTI - AGENT - 001): A mixed collective retains human and artificial agent types, protocols, resources, failures and oversight.
- **Collective decision** (SFT - SOCIAL - COLLECTIVE - DECISION - 001): Collective decision retains eligibility, options, information, rule, participation, dissent and implementation.

- **Aggregation and ecological inference boundary** (SFT - SOCIAL - AGGREGATION - LOSS - 001): Aggregation explicitly records the many-to-one loss preventing recovery of individual states from totals.

History, intervention, replication and ethics

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Historical social event** (SFT - SOCIAL - HISTORICAL - EVENT - 001): A historical event is a source-custodied reconstruction retaining perspectives, gaps and competing accounts.
- **Historical contingency** (SFT - SOCIAL - CONTINGENCY - 001): Contingency retains prior alternatives, event sequence, mechanism and unresolved counterfactuals.
- **Social causal inference** (SFT - SOCIAL - CAUSAL - INFERENCE - 001): Causal inference retains assignment, comparison, confounding, spillover, attrition and target population.
- **Social intervention** (SFT - SOCIAL - INTERVENTION - 001): Intervention evidence retains target, delivery, uptake, context, comparison, harms and missing outcomes.
- **Cross-context replication** (SFT - SOCIAL - REPLICATION - 001): Replication retains populations, settings, measures, protocol deviations, adverse outcomes and transport limits.
- **Ethical and value boundary** (SFT - SOCIAL - ETHICS - BOUNDARY - 001): Ethical judgment retains affected agents, rights, harms, benefits, disagreement and power and is not relabelled an empirical law.

Disciplinary ownership and implementation handoffs

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Mathematics and graph handoff** (SFT - SOCIAL - MATH - HANDOFF - 001): Social Science supplies measured meaning; Mathematics supplies abstract structure without selecting social identity.
- **Computation and distributed-system handoff** (SFT - SOCIAL - COMPUTATION - HANDOFF - 001): Computation may simulate a declared social process but cannot make simulated behavior an observation.
- **Biology, consciousness and individual cognition handoff** (SFT - SOCIAL - BIO - CONSC - HANDOFF - 001): Individual mechanisms and experience remain upstream evidence linked to, but not replaced by, collective relations.
- **Medicine and public-health handoff** (SFT - SOCIAL - MEDICINE - PUBLIC - HEALTH - HANDOFF - 001): Medicine owns clinical state; Social Science owns collective access and distribution; public-health composition retains both.
- **Earth and environmental handoff** (SFT - SOCIAL - EARTH - HANDOFF - 001): Earth Science owns hazard; Social Science owns exposure, vulnerability, collective response and unequal impact.
- **Engineering and policy implementation handoff** (SFT - SOCIAL - ENGINEERING - HANDOFF - 001): Engineering may test implementations against social laws but applications cannot select or rewrite those laws.

Complete claim-by-claim derivation record

The following seventy-two sections are the human-readable chain. The machine packages preserve all 256 candidate rows, 256 decisions, hostile controls, source manifest, independent output, empirical receipt and engine receipt for each claim.

1. Social agent

Claim: SFT - SOCIAL - AGENT - 001 **Family:** agents_groups_identity **Exact statement:** A social agent is a bounded acting and observed carrier whose identity, capability, context and alternatives remain explicit.

Forward chain and exact carrier

The generated carrier is bounded-acting-observed-agent-carrier. Its required relation is state-action-consequence-relation. The reconstruction must retain identity-time-capability-context-and-alternatives-held. Its declared empirical boundary is agent-identification-bounded. These coordinates are not labels added after a result: they were part of the sealed

obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `bounded-acting-observed-agent-carrier`.
- **boundary** - which population, place, period and method bound the statement: `agent-identification-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `state-action-consequence-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `identity-time-capability-context-and-alternatives-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`bounded-acting-observed-agent-carrier__agent-identification-bounded__state-action-consequence-relation__identity-time-capability-context-and-alternatives-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent,

relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if a person, role, organization or artificial agent is silently substituted for another.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `UNSD-DEMOGRAPHIC-001` - transport captured; captured bytes 88501; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.
- `IPUMS-INTERNATIONAL-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- `ILOSTAT-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social agent**, the consequence is practical and exact: A social agent is a bounded acting and observed carrier whose identity, capability, context and alternatives remain explicit. A scientific claim therefore has to expose the carrier bounded-acting-observed-agent-carrier, the relation state-action-consequence-relation, the record identity-time-capability-context-and-alternatives-held, and the boundary agent-identification-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

`sha256:9aa0e286281d1e575e03574ed87996d5d10b51642fb320803a30af7d72f6f241`; independent certificate: `sha256:7f0c74c6bdca1b38f8101cb8f52eeb417ad5401cb1039e661671e0313b337136`; engine receipt: `sha256:f2f609e054ce3520a7f646913f90e57d7dc53aaaf77ba0556570d0ff27c17ebd`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

2. Social relation

Claim: SFT-SOCIAL-RELATION-001 **Family:** agents_groups_identity **Exact statement:** A social relation is a typed, directed or reciprocal interaction between identified agents in a declared context.

Forward chain and exact carrier

The generated carrier is `ordered-agent-pair-context-carrier`. Its required relation is `typed-interaction-relation`. The reconstruction must retain `participants-direction-content-time-and-context-held`. Its declared empirical boundary is `relation-observation-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-AGENT-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `ordered-agent-pair-context-carrier`.
- **boundary** - which population, place, period and method bound the statement: `relation-observation-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `typed-interaction-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `participants-direction-content-time-and-context-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`ordered-agent-pair-context-carrier__relation-observation-bounded__typed-interaction-relation__participants-direction-content-time-and-context-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if co-presence, correlation or one-sided reporting proves a reciprocal relation.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `UNSD-DEMOGRAPHIC-001` - transport `captured`; captured bytes `88501`; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.
- `IPUMS-INTERNATIONAL-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved` and `preserved`.
- `ILOSTAT-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved` and `preserved`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social relation**, the consequence is practical and exact: A social relation is a typed, directed or reciprocal interaction between identified agents in a declared context. A scientific claim therefore has to expose the carrier `ordered-agent-pair-context-carrier`, the relation `typed-interaction-relation`, the record `participants-direction-content-time-and-context-held`, and the boundary `relation-observation-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 9aeb0a35ead80c526e186c70bd144294fe43470a110ec9d38e00564c25c1a065; independent certificate: sha256: 9c461b66d85b8d19d19fa6877ced25088ab93b9e57772bfeb1f69f9c70571d12; engine receipt: sha256: 146c15af99c64010363b325bf1d0df656f6680492ca32f2fce517b83adf8b7cf. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

3. Group identity

Claim: SFT-SOCIAL-GROUP-001 **Family:** agents_groups_identity **Exact statement:** A group retains membership, boundary, purpose, entry, exit and internal disagreement rather than becoming an indivisible person.

Forward chain and exact carrier

The generated carrier is `members-boundary-purpose-carrier`. Its required relation is `membership-and-coordination-relation`. The reconstruction must retain `members-entry-exit-purpose-time-and-disagreement-held`. Its declared empirical boundary is `group-definition-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-RELATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `members-boundary-purpose-carrier`.
- **boundary** - which population, place, period and method bound the statement: `group-definition-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `membership-and-coordination-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `members-entry-exit-purpose-time-and-disagreement-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.

- **extension** - whether any fitted weight, exception or opaque oracle has been added: **no-extra-rule**.

Their ordered product gives the unique survivor:

`members-boundary-purpose-carrier__group-definition-bounded__membership-and-coordination-relation__members-entry-exit-purpose-time-and-disagreement-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is **depth_independent**. Minimality passed: **True**. Named-form uniqueness passed: **True**. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if a label or sample is treated as a coherent group without membership evidence.

- **false_premise** - passed **True**: reject the generated form lacking the complete physical carrier
- **tampered_source** - passed **True**: reject a changed registered source identity
- **tampered_artifact** - passed **True**: reject a missing, duplicate or additional survivor
- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **UNSD-DEMOGRAPHIC-001** - transport **captured**; captured bytes **88501**; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.
- **IPUMS-INTERNATIONAL-001** - transport **failed_preserved**; captured bytes **0**; snapshot **unresolved and preserved**.
- **ILOSTAT-001** - transport **failed_preserved**; captured bytes **0**; snapshot **unresolved and preserved**.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **group identity**, the consequence is practical and exact: A group retains membership, boundary, purpose, entry, exit and internal disagreement rather than becoming an indivisible person. A scientific claim therefore has to expose the carrier `members-boundary-purpose-carrier`, the relation `membership-and-coordination-relation`, the record `members-entry-exit-purpose-time-and-disagreement-held`, and the boundary

group-definition-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:3d94b6fca6fa1a8a0da49d56d63ff1374029f33e773144ad683bd15369d227ac; independent certificate: sha256:e916c9cca85a24f571f5df78b18f2274cd5a12ef6091e1e292c0662c2d203266; engine receipt: sha256:ea38d831ac38a601d2aaedcfa5153d810f8ca7d31b6227d420da9b8c6956f387. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

4. Collective state

Claim: SFT-SOCIAL-COLLECTIVE-001 **Family:** agents_groups_identity **Exact statement:** A collective state composes agent states and relations while retaining internal distribution and minority states.

Forward chain and exact carrier

The generated carrier is **agents-relations-shared-state-carrier**. Its required relation is **joint-state-composition-relation**. The reconstruction must retain **members-links-distribution-time-and-minority-states-held**. Its declared empirical boundary is **collective-observation-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GROUP-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: **agents-relations-shared-state-carrier**.
- **boundary** - which population, place, period and method bound the statement: **collective-observation-bounded**.
- **relation** - which generated interaction or comparison connects the carriers: **joint-state-composition-relation**.

- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
members-links-distribution-time-and-minority-states-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

agents-relations-shared-state-carrier__collective-observation-bounded__joint-state-composition-relation__members-links-distribution-time-and-minority-states-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if an aggregate erases all constituent variation or becomes an individual mind.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `UNSD-DEMOGRAPHIC-001` - transport captured; captured bytes 88501; snapshot sha256: cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.
- `IPUMS-INTERNATIONAL-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- `ILOSTAT-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not

claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective state**, the consequence is practical and exact: A collective state composes agent states and relations while retaining internal distribution and minority states. A scientific claim therefore has to expose the carrier **agents-relations-shared-state-carrier**, the relation **joint-state-composition-relation**, the record **members-links-distribution-time-and-minority-states-held**, and the boundary **collective-observation-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:1e213b8e450f2ca5fd440a09b4b06614593d6db3840afb5c2fd3f8d232c025f9; independent certificate: sha256:5759ffd91cce60f60a36ee94cd86abec93509cccca90880d27c1f398ff9a8063; engine receipt: sha256:f0866c38e5fc8d9c9dc0190ddb5a8994de2b679bddf4406a43141629c5ee9cda. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

5. Organization identity

Claim: SFT-SOCIAL-ORGANIZATION-001 **Family:** agents_groups_identity **Exact statement:** An organization is a versioned arrangement of members, roles, rules, resources and processes.

Forward chain and exact carrier

The generated carrier is **roles-members-rules-resources-carrier**. Its required relation is **role-and-process-organization**. The reconstruction must retain **membership-authority-rules-resources-time-and-version-held**. Its declared empirical boundary is **organization-boundary-declared**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `roles-members-rules-resources-carrier`.
- **boundary** - which population, place, period and method bound the statement: `organization-boundary-declared`.
- **relation** - which generated interaction or comparison connects the carriers: `role-and-process-organization`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `membership-authority-rules-resources-time-and-version-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`roles-members-rules-resources-carrier__organization-boundary-declared__role-and-process-organization__membership-authority-rules-resources-time-and-version-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if a legal name alone establishes stable organization or unchanged purpose.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- **UNSD-DEMOGRAPHIC-001** - transport captured; captured bytes 88501; snapshot sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.
- **IPUMS-INTERNATIONAL-001** - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- **ILOSTAT-001** - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **organization identity**, the consequence is practical and exact: An organization is a versioned arrangement of members, roles, rules, resources and processes. A scientific claim therefore has to expose the carrier roles-members-rules-resources-carrier, the relation role-and-process-organization, the record membership-authority-rules-resources-time-and-version-held, and the boundary organization-boundary-declared together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:b51f090e2e0186c60ae25ee38e3d8a67f989a0feab7d1b8623fff59970ee93d4; independent certificate: sha256:a360f569326af227f28930803909d1950580ff649c67ab4d521f929a620dac65; engine receipt: sha256:422c2814a0efc03176f88be51639b782eb4bf45afb9cd4dbf9bdd83660547e9e. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

6. Social population

Claim: SFT-SOCIAL-POPULATION-001 **Family:** agents_groups_identity **Exact statement:** A social population is a sampling-frame-bound agent set with selection, weighting, censoring and missingness retained.

Forward chain and exact carrier

The generated carrier is selection-bounded-agent-set-carrier. Its required relation is membership-distribution-and-weighting-relation. The reconstruction must retain frame-selection-time-weights-censoring-and-missingness-held. Its declared empirical boundary is population-sampling-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001,

SFT - SOCIAL - ORGANIZATION - 001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `selection-bounded-agent-set-carrier`.
- **boundary** - which population, place, period and method bound the statement: `population-sampling-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `membership-distribution-and-weighting-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `frame-selection-time-weights-censoring-and-missingness-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
selection-bounded-agent-set-carrier__population-sampling-bounded__membership-distribution-and-weighting-relation__frame-selection-time-weights-censoring-and-missingness-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject universal inference from favorable respondents or an undefined denominator.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `UNSD-DEMOGRAPHIC-001` - transport captured; captured bytes 88501; snapshot sha256: cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.
- `IPUMS-INTERNATIONAL-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- `ILOSTAT-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social population**, the consequence is practical and exact: A social population is a sampling-frame-bound agent set with selection, weighting, censoring and missingness retained. A scientific claim therefore has to expose the carrier `selection-bounded-agent-set-carrier`, the relation `membership-distribution-and-weighting-relation`, the record `frame-selection-time-weights-censoring-and-missingness-held`, and the boundary `population-sampling-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256: 9bbb9038c173f4f3295f99cf90335d41bc8b21f4f7c359a2eb68300046d279c8; independent certificate: sha256: c9e4cf25500edf37bde6e3b927a773ebc772182edbb4b05f978467cb7e32e401; engine receipt: sha256: 176f9f0a8f0c176a6beca49d36d60c4310ba772a229681fc2eb7a7c3f6187ece. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

7. Social observation

Claim: `SFT-SOCIAL-OBSERVATION-001` **Family:** `observation_action_communication` **Exact statement:** Social observation retains observer, method, consent, setting, reactivity and unavailable records.

Forward chain and exact carrier

The generated carrier is `observer-agent-setting-record-carrier`. Its required relation is `observation-and-reactivity-relation`. The reconstruction must retain `observer-method-consent-setting-time-and-reactivity-held`. Its declared empirical boundary is `observation-method-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-POPULATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `observer-agent-setting-record-carrier`.
- **boundary** - which population, place, period and method bound the statement: `observation-method-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `observation-and-reactivity-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `observer-method-consent-setting-time-and-reactivity-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`observer-agent-setting-record-carrier__observation-method-bounded__observation-and-reactivity-relation__observer-method-consent-setting-time-and-reactivity-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if observation is treated as intervention-free or private inference as direct report.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `GSS-NORC-001` - transport captured; captured bytes 67309; snapshot `sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1`.
- `ICPSR-DATA-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- `IPUMS-INTERNATIONAL-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social observation**, the consequence is practical and exact: Social observation retains observer, method, consent, setting, reactivity and unavailable records. A scientific claim therefore has to expose the carrier `observer-agent-setting-record-carrier`, the relation `observation-and-reactivity-relation`, the record `observer-method-consent-setting-time-and-reactivity-held`, and the boundary `observation-method-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

`sha256:0b0f6e6e5248c46d820a7322e33a4fee588ed491c70574c283f3b627ffa4f3de`; independent certificate: `sha256:9a174fd7e40a98391357df6582c5ecf55eeb5ecaa9ba8672bfec06901b367377`; engine receipt: `sha256:9e5f48e91a6767b794854f5c461addb7d9016687a4721cc0970c23fa475f47a6`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

8. Social action

Claim: `SFT-SOCIAL-ACTION-001` **Family:** `observation_action_communication` **Exact statement:** Action attribution retains capability, alternatives, constraints, intent evidence, act and consequence.

Forward chain and exact carrier

The generated carrier is **agent-choice-context-consequence-carrier**. Its required relation is **capability-to-action-relation**. The reconstruction must retain **alternatives-constraints-intent-act-outcome-and-time-held**. Its declared empirical boundary is **action-attribution-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-POPULATION-001, SFT-SOCIAL-OBSERVATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: **agent-choice-context-consequence-carrier**.
- **boundary** - which population, place, period and method bound the statement: **action-attribution-bounded**.
- **relation** - which generated interaction or comparison connects the carriers: **capability-to-action-relation**.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: **alternatives-constraints-intent-act-outcome-and-time-held**.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: **evidence-and-normative-class-explicit**.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: **root-bound-forward-forcing**.
- **generality** - whether finite extension preserves every context rather than universalising one case: **positive-finite-extension-retains-all-contexts**.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: **no-extra-rule**.

Their ordered product gives the unique survivor:

agent-choice-context-consequence-carrier__action-attribution-bounded__capability-to-action-relation__alternatives-constraints-intent-act-outcome-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model

or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if outcome alone proves intent or if constrained conduct is treated as free choice.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social action**, the consequence is practical and exact: Action attribution retains capability, alternatives, constraints, intent evidence, act and consequence. A scientific claim therefore has to expose the carrier `agent-choice-context-consequence-carrier`, the relation `capability-to-action-relation`, the record `alternatives-constraints-intent-act-outcome-and-time-held`, and the boundary `action-attribution-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 9d8810a1edd01cb9a641eee2f25490238fd37c8c63e47fa7918e62b046349410; independent

certificate: sha256:110df5fbd71c4b1284afc207d5fa3d8e63ede169d8f091573229f72401285d2d; engine receipt: sha256:99446d01ce954f8802de92a9839ef59588352022b9b9b8665cc6a5ecd70d0c82. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

9. Communication event

Claim: SFT-SOCIAL-MESSAGE-001 **Family:** observation_action_communication **Exact statement:** A communication event retains sender, content, channel, recipient opportunity, reception and loss.

Forward chain and exact carrier

The generated carrier is sender-message-channel-recipient-carrier. Its required relation is encoding-transmission-reception-relation. The reconstruction must retain source-content-channel-time-reach-and-loss-held. Its declared empirical boundary is communication-event-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-POPULATION-001, SFT-SOCIAL-ACTION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: sender-message-channel-recipient-carrier.
- **boundary** - which population, place, period and method bound the statement: communication-event-bounded.
- **relation** - which generated interaction or comparison connects the carriers: encoding-transmission-reception-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: source-content-channel-time-reach-and-loss-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

sender-message-channel-recipient-carrier__communication-event-bounded__encoding-transmission-reception-relation__source-content-channel-time-reach-and-loss-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retain-s-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is **depth_independent**. Minimality passed: **True**. Named-form uniqueness passed: **True**. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if sending proves receipt, receipt proves belief, or exposure proves action.

- **false_premise** - passed **True**: reject the generated form lacking the complete physical carrier
- **tampered_source** - passed **True**: reject a changed registered source identity
- **tampered_artifact** - passed **True**: reject a missing, duplicate or additional survivor
- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: **source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence**. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **GSS-NORC-001** - transport **captured**; captured bytes 67309; snapshot sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- **ICPSR-DATA-001** - transport **failed_preserved**; captured bytes 0; snapshot **unresolved** and **preserved**.
- **IPUMS-INTERNATIONAL-001** - transport **failed_preserved**; captured bytes 0; snapshot **unresolved** and **preserved**.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **communication event**, the consequence is practical and exact: A communication event retains sender, content, channel, recipient opportunity, reception and loss. A scientific claim therefore has to expose the carrier

sender-message-channel-recipient-carrier, the relation encoding-transmission-reception-relation, the record

source-content-channel-time-reach-and-loss-held, and the boundary communication-event-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application

choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:1116cbb366ffcb0be538cf43a2bb7d616432e01edfed1dad0d81508a02afdc44; independent certificate: sha256:5cedd8e07c5660ebef3b556d9095066a77608675951c6256b68fb1d48324ef8e; engine receipt: sha256:f3674a401671a9f62232868be604578e8d23a1b629982ac17a0a58250f24450e. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

10. Shared social record

Claim: SFT-SOCIAL-SHARED-RECORD-001 **Family:** observation_action_communication **Exact statement:** A shared record is a versioned artifact with authorship, access, custody, changes and disputes retained.

Forward chain and exact carrier

The generated carrier is `participants-artifact-access-carrier`. Its required relation is `write-read-and-custody-relation`. The reconstruction must retain `authors-version-access-changes-and-disputes-held`. Its declared empirical boundary is `record-custody-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-POPULATION-001, SFT-SOCIAL-MESSAGE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `participants-artifact-access-carrier`.
- **boundary** - which population, place, period and method bound the statement: `record-custody-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `write-read-and-custody-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `authors-version-access-changes-and-disputes-held`.

- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

participants-artifact-access-carrier__record-custody-bounded__write-read-and-custody-relation__authors-version-access-changes-and-disputes-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if one party's record becomes uncontested collective memory.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `GSS-NORC-001` - transport `captured`; captured bytes `67309`; snapshot `sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1`.
- `ICPSR-DATA-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.
- `IPUMS-INTERNATIONAL-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and

adverse rows retained.

Scientific meaning

For **shared social record**, the consequence is practical and exact: A shared record is a versioned artifact with authorship, access, custody, changes and disputes retained. A scientific claim therefore has to expose the carrier `participants-artifact-access-carrier`, the relation `write-read-and-custody-relation`, the record `authors-version-access-changes-and-disputes-held`, and the boundary `record-custody-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:b662dc9f66a39e517078e882fedbbd96a05b4361aea3be4b7fc90f6d4e70db98; independent certificate: sha256:273d068dfdfefb94674679f07f3eeb587aeba13a8fb5095d266dc91dbaede210b; engine receipt: sha256:f56e1e73f396479164cdf9c2e4688d84f875d812d4f97c17ce23efbe47ae6ee1. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

11. Common-knowledge boundary

Claim: SFT-SOCIAL-COMMON-KNOWLEDGE-001 **Family:** `observation_action_communication` **Exact statement:** Common knowledge claims require a declared finite mutual-observation depth and complete receipt chain.

Forward chain and exact carrier

The generated carrier is `agents-message-receipt-iteration-carrier`. Its required relation is `mutual-observation-depth-relation`. The reconstruction must retain `participants-receipts-depth-and-missing-links-held`. Its declared empirical boundary is `finite-knowledge-depth-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-POPULATION-001, SFT-SOCIAL-SHARED-RECORD-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-message-receipt-iteration-carrier`.
- **boundary** - which population, place, period and method bound the statement: `finite-knowledge-depth-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `mutual-observation-depth-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `participants-receipts-depth-and-missing-links-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`agents-message-receipt-iteration-carrier__finite-knowledge-depth-bounded__mutual-observation-depth-relation__participants-receipts-depth-and-missing-links-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject completed infinite mutual knowledge or infer it from a public message alone.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **common-knowledge boundary**, the consequence is practical and exact: Common knowledge claims require a declared finite mutual-observation depth and complete receipt chain. A scientific claim therefore has to expose the carrier agents-message-receipt-iteration-carrier, the relation mutual-observation-depth-relation, the record participants-receipts-depth-and-missing-links-held, and the boundary finite-knowledge-depth-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:71c761cd89d0205dffbc921a7e7697062312d8b84cb6b25b7f0f08a5e0796ef; independent certificate: sha256:c4ac04442a2ec7b398bdd2319648c25102ffdb8fce90f7507b6f496d5eb96c35; engine receipt: sha256:3b5736ee7fbcd85b03e2edad2e4b2a5447fb2baa06d55f9e704e3b01eb099fbd. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

12. Social communication noise

Claim: SFT-SOCIAL-COMMUNICATION-NOISE-001 **Family:** observation_action_communication **Exact statement:** Communication noise retains technical loss, ambiguity, reinterpretation and audience differences.

Forward chain and exact carrier

The generated carrier is message-channel-context-interpretation-carrier. Its required relation is distortion-ambiguity-and-loss-relation. The reconstruction must retain source-versions-audiences-interpretations-and-missingness-held. Its declared empirical boundary is channel-and-audience-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-POPULATION-001,

SFT - SOCIAL - COMMON - KNOWLEDGE - 001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `message-channel-context-interpretation-carrier`.
- **boundary** - which population, place, period and method bound the statement: `channel-and-audience-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `distortion-ambiguity-and-loss-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `source-versions-audiences-interpretations-and-missingness-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
message-channel-context-interpretation-carrier__channel-and-audience-bounded__distortion-ambiguity-and-loss-relation__source-versions-audiences-interpretations-and-missingness-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if disagreement is automatically deception or if a transformed message is called identical.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social communication noise**, the consequence is practical and exact: Communication noise retains technical loss, ambiguity, reinterpretation and audience differences. A scientific claim therefore has to expose the carrier `message-channel-context-interpretation-carrier`, the relation `distortion-ambiguity-and-loss-relation`, the record `source-versions-audiences-interpretations-and-missingness-held`, and the boundary `channel-and-audience-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256: e5650bb5680cf1234de2c1018357e47154abe126baf1ad0e5b52e9c2fcf55936; independent certificate: sha256: adfb6dae57e10532692d670594936c5fe7d24ac4154991590103eb8f5784c412; engine receipt: sha256: c609ce1483a76a7a83d8f28c5fd7170640788f6afeff041858b81e20aa23150e. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

13. Coordination

Claim: SFT-SOCIAL-COORDINATION-001 **Family:** `coordination_conflict` **Exact statement:** Coordination is compatible action around a declared target with dependencies, timing and failures retained.

Forward chain and exact carrier

The generated carrier is `agents-actions-joint-target-carrier`. Its required relation is `timing-and-compatibility-relation`. The reconstruction must retain `agents-actions-dependencies-time-failures-and-target-held`. Its declared empirical boundary is `task-and-context-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COMMUNICATION-NOISE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-actions-joint-target-carrier`.
- **boundary** - which population, place, period and method bound the statement: `task-and-context-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `timing-and-compatibility-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `agents-actions-dependencies-time-failures-and-target-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`agents-actions-joint-target-carrier__task-and-context-bounded__timing-and-compatibility-relation__agents-actions-dependencies-time-failures-and-target-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent,

relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if simultaneous action proves a shared target or successful coordination proves cooperation.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- `UCDP-DOWNLOADS-001` - transport captured; captured bytes 138078; snapshot sha256: 5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- `UNODC-DATA-001` - transport captured; captured bytes 44411; snapshot sha256: 35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- `GSS-NORC-001` - transport captured; captured bytes 67309; snapshot sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **coordination**, the consequence is practical and exact: Coordination is compatible action around a declared target with dependencies, timing and failures retained. A scientific claim therefore has to expose the carrier `agents-actions-joint-target-carrier`, the relation `timing-and-compatibility-relation`, the record `agents-actions-dependencies-time-failures-and-target-held`, and the boundary `task-and-context-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 8ee843badaeca2dffe17716651d75ccdf841d85d07ae08290d49ce1b84645d3c; independent certificate: sha256: ce5862b80b25720fd9bb1047054e0cfd2ac7764da3c1167334ca71d426858e4f; engine receipt: sha256: 332c7b9b943206b9afa4b5b83ce4fc3c08cf81653588e2c217d2966cfc0fcd6d. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

14. Cooperation

Claim: SFT-SOCIAL-COOPERATION-001 **Family:** coordination_conflict **Exact statement:** Cooperation retains contributions, costs, benefits, alternatives and defection for identified participants.

Forward chain and exact carrier

The generated carrier is `agents-contributions-joint-benefit-carrier`. Its required relation is `contribution-benefit-and-reciprocity-relation`. The reconstruction must retain `contributions-costs-benefits-alternatives-time-and-defection-held`. Its declared empirical boundary is `benefit-and-membership-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COMMUNICATION-NOISE-001, SFT-SOCIAL-COORDINATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-contributions-joint-benefit-carrier`.
- **boundary** - which population, place, period and method bound the statement: `benefit-and-membership-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `contribution-benefit-and-reciprocity-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `contributions-costs-benefits-alternatives-time-and-defection-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

agents-contributions-joint-benefit-carrier__benefit-and-membership-bounded__contribution-benefit-and-reciprocity-relation__contributions-costs-benefits-alternatives-time-and-defection-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if aggregate gain hides uncompensated loss or coerced contribution.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `UCDP-DOWNLOADS-001` - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- `UNODC-DATA-001` - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- `GSS-NORC-001` - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **cooperation**, the consequence is practical and exact: Cooperation retains contributions, costs, benefits, alternatives and defection for identified participants. A scientific claim therefore has to expose the carrier

`agents-contributions-joint-benefit-carrier`, the relation `contribution-benefit-and-reciprocity-relation`, the record `contributions-costs-benefits-alternatives-time-and-defection-held`, and the boundary `benefit-and-membership-bounded` together. If any one is hidden, the result may remain a description, interpretation,

model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:dcc4601ab91b90960bb694d29ee3f425a79d8f621ec6a36789dc81c4144e624e; independent certificate: sha256:94066ec99ceaa3ad4b9bb8881f2fc59f9008a5bc906070f773c64e196e5bde9d; engine receipt: sha256:5c58d6b54a1282c18b43b1785af6f6f7689a713dee415eb607df14171fc37ea5. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

15. Competition

Claim: SFT-SOCIAL-COMPETITION-001 **Family:** coordination_conflict **Exact statement:** Competition is a rule- and resource-bounded rival claim with opportunities and exclusion retained.

Forward chain and exact carrier

The generated carrier is `agents-scarce-target-rules-carrier`. Its required relation is `rival-claim-and-allocation-relation`. The reconstruction must retain `participants-resource-rules-opportunities-outcomes-and-exclusion-held`. Its declared empirical boundary is `contest-boundary-declared`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COMMUNICATION-NOISE-001, SFT-SOCIAL-COOPERATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-scarce-target-rules-carrier`.
- **boundary** - which population, place, period and method bound the statement: `contest-boundary-declared`.
- **relation** - which generated interaction or comparison connects the carriers: `rival-claim-and-allocation-relation`.

- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
participants-resource-rules-opportunities-outcomes-and-exclusion-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

agents-scarce-target-rules-carrier__contest-boundary-declared__rival-claim-and-allocation-relation__participants-resource-rules-opportunities-outcomes-and-exclusion-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if unequal access is hidden or all parallel striving is called competition.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256: 5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256: 35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **competition**, the consequence is practical and exact: Competition is a rule- and resource-bounded rival claim with opportunities and exclusion retained. A scientific claim therefore has to expose the carrier **agents-scarce-target-rules-carrier**, the relation **rival-claim-and-allocation-relation**, the record **participants-resource-rules-opportunities-outcomes-and-exclusion-held**, and the boundary **contest-boundary-declared** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 85ff9b4dc5a0592bd60ec5457f5d1887210327d677436de07084eb6e53037dfb; independent certificate: sha256: 025c6b194e0454aca243c1388f5407a90dadf9183ea76cc2b0f8697af9b3e0de; engine receipt: sha256: 6b6d00922b1c86a7c0c38187646d9ec0c62b74075afe7fd0a9b05cc064d6488b. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

16. Conflict

Claim: SFT-SOCIAL-CONFLICT-001 **Family:** coordination_conflict **Exact statement:** Conflict retains incompatible claims, capabilities, actions, harms, alternatives and time.

Forward chain and exact carrier

The generated carrier is **agents-incompatible-claims-action-carrier**. Its required relation is **claim-opposition-and-harm-relation**. The reconstruction must retain **claims-capabilities-actions-harms-time-and-alternatives-held**. Its declared empirical boundary is **conflict-definition-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COMMUNICATION-NOISE-001, SFT-SOCIAL-COMPETITION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered

preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-incompatible-claims-action-carrier`.
- **boundary** - which population, place, period and method bound the statement: `conflict-definition-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `claim-opposition-and-harm-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `claims-capabilities-actions-harms-time-and-alternatives-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
agents-incompatible-claims-action-carrier__conflict-definition-bounded__claim-oppositio
n-and-harm-relation__claims-capabilities-actions-harms-time-and-alternatives-held__evid
ence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extensio
n-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if disagreement alone becomes violence or one source defines every party's claim.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0.

The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **conflict**, the consequence is practical and exact: Conflict retains incompatible claims, capabilities, actions, harms, alternatives and time. A scientific claim therefore has to expose the carrier `agents-incompatible-claims-action-carrier`, the relation `claim-opposition-and-harm-relation`, the record `claims-capabilities-actions-harms-time-and-alternatives-held`, and the boundary `conflict-definition-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:9ebd08ecabf8024f5ef9fdf83370929ddef651fa0d919d8571edd631a1f0f470; independent
certificate: sha256:2c810b7e41b0cc35b593d40f4cd19904d1a73bfc411618227df50f53f8ff05c4; engine
receipt: sha256:0c2dfd4f8e5ce16ff07f28527f9e08bb0d4677836a294403856f3dc50d04780. The
independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

17. Bargaining

Claim: SFT-SOCIAL-BARGAINING-001 **Family:** `coordination_conflict` **Exact statement:** Bargaining retains offers, outside options, information, power, time, breakdown and agreement.

Forward chain and exact carrier

The generated carrier is `parties-offers-options-agreement-carrier`. Its required relation is `offer-counteroffer-and-acceptance-relation`. The reconstruction must retain `options-information-power-time-breakdown-and-outcome-held`. Its declared empirical boundary is `bargaining-episode-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001,
SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001,
SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001,
SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001,

SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001,
 SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001,
 SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001,
 SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001,
 SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001,
 SFT-SOCIAL-COMMUNICATION-NOISE-001, SFT-SOCIAL-CONFLICT-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `parties-offers-options-agreement-carrier`.
- **boundary** - which population, place, period and method bound the statement: `bargaining-episode-bounded`.
- **relation** - which generated interaction or comparison connects the carriers:
`offer-counteroffer-and-acceptance-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
`options-information-power-time-breakdown-and-outcome-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
`evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
`root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case:
`positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`parties-offers-options-agreement-carrier__bargaining-episode-bounded__offer-counteroffer-and-acceptance-relation__options-information-power-time-breakdown-and-outcome-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if the final agreement proves equal choice or erased alternatives.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier

- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **bargaining**, the consequence is practical and exact: Bargaining retains offers, outside options, information, power, time, breakdown and agreement. A scientific claim therefore has to expose the carrier `parties-offers-options-agreement-carrier`, the relation `offer-counteroffer-and-acceptance-relation`, the record `options-information-power-time-breakdown-and-outcome-held`, and the boundary `bargaining-episode-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256:ba594abcde4302b003da60c65d7173467f75f63e1e2b72d60cd8a640758b337d; independent
certificate: sha256:74934c9c5d0c97738e500d38d0ded2577a73903b8b6d62bf9f1ec401eada9266; engine
receipt: sha256:b452e58367879549904f8cc507672a5ef1bbae7e43052b92ec7365f31d52ead9. The
independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

18. Collective action

Claim: SFT-SOCIAL-COLLECTIVE-ACTION-001 **Family:** `coordination_conflict` **Exact statement:** Collective action retains participation, abstention, contribution, threshold, benefit, failure and context.

Forward chain and exact carrier

The generated carrier is `agents-contribution-threshold-outcome-carrier`. Its required relation is `participation-and-threshold-relation`. The reconstruction must retain `participants-nonparticipants-costs-benefits-time-and-failures-held`. Its declared empirical boundary is `action-frame-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-COMMUNICATION-NOISE-001`, `SFT-SOCIAL-BARGAINING-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-contribution-threshold-outcome-carrier`.
- **boundary** - which population, place, period and method bound the statement: `action-frame-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `participation-and-threshold-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `participants-nonparticipants-costs-benefits-time-and-failures-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`agents-contribution-threshold-outcome-carrier__action-frame-bounded__participation-and-threshold-relation__participants-nonparticipants-costs-benefits-time-and-failures-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the

result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if outcome visibility erases nonparticipants or failed attempts.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `UCDP-DOWNLOADS-001` - transport captured; captured bytes 138078; snapshot `sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5`.
- `UNODC-DATA-001` - transport captured; captured bytes 44411; snapshot `sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57`.
- `GSS-NORC-001` - transport captured; captured bytes 67309; snapshot `sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective action**, the consequence is practical and exact: Collective action retains participation, abstention, contribution, threshold, benefit, failure and context. A scientific claim therefore has to expose the carrier `agents-contribution-threshold-outcome-carrier`, the relation `participation-and-threshold-relation`, the record `participants-nonparticipants-costs-benefits-time-and-failures-held`, and the boundary `action-frame-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 31250970e3490d01e0e402efd510a149f19999abe12c57de4b27fa6fecc62508; independent certificate: sha256: fb3ab7f0710db8a3010a39b62f683894e74c6f3e3f546b8aac8ba588c770e837; engine receipt: sha256: a7185420f280dd7879b17b1b675bcbdb8920bf7d9bcd752b60733ccf468ae2760. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

19. Social norm

Claim: SFT-SOCIAL-NORM-001 **Family:** norm_institution_enforcement **Exact statement:** A norm is a context-bound expectation evidenced by behavior and responses to conformity and violation.

Forward chain and exact carrier

The generated carrier is `group-expectation-behaviour-response-carrier`. Its required relation is `expectation-conformity-and-sanction-relation`. The reconstruction must retain `group-context-variation-violations-responses-and-time-held`. Its declared empirical boundary is `norm-population-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-ACTION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `group-expectation-behaviour-response-carrier`.
- **boundary** - which population, place, period and method bound the statement: `norm-population-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `expectation-conformity-and-sanction-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `group-context-variation-violations-responses-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.

- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

group-expectation-behaviour-response-carrier__norm-population-bounded__expectation-conformity-and-sanction-relation__group-context-variation-violations-responses-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if researcher preference or majority behavior alone becomes a norm.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `VDEM-DATA-001` - transport captured; captured bytes 122924; snapshot
sha256: cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- `UNODC-DATA-001` - transport captured; captured bytes 44411; snapshot
sha256: 35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- `OECD-DATA-EXPLORER-001` - transport captured; captured bytes 173987; snapshot
sha256: cdac198c1032dbb902cd9bfd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social norm**, the consequence is practical and exact: A norm is a context-bound expectation evidenced by behavior and responses to conformity and violation. A scientific claim therefore has to expose the carrier `group-expectation-behaviour-response-carrier`, the relation `expectation-conformity-and-sanction-relation`, the record `group-context-variation-violations-responses-and-time-held`, and the boundary `norm-population-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:cf47c5430671a2ee6d7877ec0a093e0831416bef79d484252710974f8cec2ad8; independent certificate: sha256:c55672eb76bdefc2f65911803490cc866b4b17446ce048dbd955991296ef1b2c; engine receipt: sha256:81ed8141611660faf49244ba772291a6a232970c571f398f99fd19b9bf11bf7b. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

20. Social rule

Claim: SFT-SOCIAL-RULE-001 **Family:** norm_institution_enforcement **Exact statement:** A rule retains issuer, text, version, scope, subjects, exceptions, application and contestation.

Forward chain and exact carrier

The generated carrier is `issuer-text-scope-subject-carrier`. Its required relation is `rule-application-relation`. The reconstruction must retain `authority-text-version-scope-exceptions-and-contestation-held`. Its declared empirical boundary is `rule-jurisdiction-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-ACTION-001, SFT-SOCIAL-NORM-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `issuer-text-scope-subject-carrier`.
- **boundary** - which population, place, period and method bound the statement: `rule-jurisdiction-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `rule-application-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `authority-text-version-scope-exceptions-and-contestation-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
issuer-text-scope-subject-carrier__rule-jurisdiction-bounded__rule-application-relation
__authority-text-version-scope-exceptions-and-contestation-held__evidence-and-normative
-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-cont
exts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if an unwritten inference is presented as a universal rule or version changes disappear.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `VDEM-DATA-001` - transport captured; captured bytes 122924; snapshot `sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c`.

- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- OECD-DATA-EXPLORER-001 - transport captured; captured bytes 173987; snapshot
sha256:cdac198c1032dbb902cd9bfd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social rule**, the consequence is practical and exact: A rule retains issuer, text, version, scope, subjects, exceptions, application and contestation. A scientific claim therefore has to expose the carrier **issuer-text-scope-subject-carrier**, the relation **rule-application-relation**, the record **authority-text-version-scope-exceptions-and-contestation-held**, and the boundary **rule-jurisdiction-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:1d63d026129a48e15641d759f95b78b90f6fa9cbf102bf7886bb6181f22c45ff; independent certificate: sha256:db327835791bf997a0e987ab45de704aee7fa5d8159d0900863b26fa09f55eaa; engine receipt: sha256:e7ed142368911acc1f16f02f17d142061e03ea2114d5c3afa799e2345403f4b7. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

21. Institution

Claim: SFT-SOCIAL-INSTITUTION-001 **Family:** norm_institution_enforcement **Exact statement:** An institution is a historically repeated governed process, not merely a building, name or consensus.

Forward chain and exact carrier

The generated carrier is **roles-rules-practices-resources-carrier**. Its required relation is **repeated-governed-process-relation**. The reconstruction must retain **history-membership-rules-resources-compliance-and-change-held**. Its declared empirical boundary is **institution-context-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-ACTION-001, SFT-SOCIAL-RULE-001. Each dependency was model-admitted before this

claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `roles-rules-practices-resources-carrier`.
- **boundary** - which population, place, period and method bound the statement: `institution-context-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `repeated-governed-process-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `history-membership-rules-resources-compliance-and-change-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
roles-rules-practices-resources-carrier__institution-context-bounded__repeated-governed-
-process-relation__history-membership-rules-resources-compliance-and-change-held__evidence-
and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-
retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if institutional prestige is used as evidence or if internal conflict is erased.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- OECD-DATA-EXPLORER-001 - transport captured; captured bytes 173987; snapshot
sha256:cdac198c1032dbb902cd9bfd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **institution**, the consequence is practical and exact: An institution is a historically repeated governed process, not merely a building, name or consensus. A scientific claim therefore has to expose the carrier `roles-rules-practices-resources-carrier`, the relation `repeated-governed-process-relation`, the record `history-membership-rules-resources-compliance-and-change-held`, and the boundary `institution-context-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:368c7fa5c901149bd3a9a726af0648c199377412b11bf2b69e1f442255f2bbda; independent certificate: sha256:e81c8c521bf87c0d117c847bf370e1ae7cad09cea9da56a286d1676a1aa505fb; engine receipt: sha256:b9e33b4de5564358ded810f93b3298bea090786567f936270ba09ba3569c9918. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

22. Enforcement

Claim: SFT-SOCIAL-ENFORCEMENT-001 **Family:** `norm_institution_enforcement` **Exact statement:** Enforcement retains monitoring, selectivity, procedure, response, appeal and non-enforcement.

Forward chain and exact carrier

The generated carrier is `rule-monitor-response-subject-carrier`. Its required relation is `detection-to-response-relation`. The reconstruction must retain `authority-monitoring-selectivity-process-response-and-appeal-held`. Its declared empirical boundary is `enforcement-regime-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001,

SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-ACTION-001, SFT-SOCIAL-INSTITUTION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `rule-monitor-response-subject-carrier`.
- **boundary** - which population, place, period and method bound the statement: `enforcement-regime-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `detection-to-response-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `authority-monitoring-selectivity-process-response-and-appeal-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`rule-monitor-response-subject-carrier__enforcement-regime-bounded__detection-to-response-relation__authority-monitoring-selectivity-process-response-and-appeal-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if a written rule proves uniform enforcement or punishment proves legitimacy.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `VDEM-DATA-001` - transport captured; captured bytes 122924; snapshot
sha256: cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- `UNODC-DATA-001` - transport captured; captured bytes 44411; snapshot
sha256: 35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- `OECD-DATA-EXPLORER-001` - transport captured; captured bytes 173987; snapshot
sha256: cdac198c1032dbb902cd9bfd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **enforcement**, the consequence is practical and exact: Enforcement retains monitoring, selectivity, procedure, response, appeal and non-enforcement. A scientific claim therefore has to expose the carrier `rule-monitor-response-subject-carrier`, the relation `detection-to-response-relation`, the record `authority-monitoring-selectivity-process-response-and-appeal-held`, and the boundary `enforcement-regime-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: e0842d49fdb0a2111b5efa85a79362a99a4bfffecfdec29ea2ac2aceac342dc31; independent certificate: sha256: f55ac6ca28e96c56b55e84ba317644a10a3f61ac14939cd93a119a14a1cdcd9f; engine receipt: sha256: b998039d03a7d316d7dbfea6e71a7b83d5a0774f84727d8b47839b3c174a84f2. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

23. Compliance and resistance

Claim: `SFT-SOCIAL-COMPLIANCE-001` **Family:** `norm_institution_enforcement` **Exact statement:** Compliance and resistance remain distinct from ignorance, inability, concealment and coincidence.

Forward chain and exact carrier

The generated carrier is **subject-rule-action-reason-carrier**. Its required relation is **rule-to-behaviour-comparison**. The reconstruction must retain **knowledge-capability-motive-action-resistance-and-consequence-held**. Its declared empirical boundary is **compliance-observation-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-ACTION-001, SFT-SOCIAL-ENFORCEMENT-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: **subject-rule-action-reason-carrier**.
- **boundary** - which population, place, period and method bound the statement: **compliance-observation-bounded**.
- **relation** - which generated interaction or comparison connects the carriers: **rule-to-behaviour-comparison**.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: **knowledge-capability-motive-action-resistance-and-consequence-held**.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: **evidence-and-normative-class-explicit**.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: **root-bound-forward-forcing**.
- **generality** - whether finite extension preserves every context rather than universalising one case: **positive-finite-extension-retains-all-contexts**.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: **no-extra-rule**.

Their ordered product gives the unique survivor:

subject-rule-action-reason-carrier__compliance-observation-bounded__rule-to-behaviour-comparison__knowledge-capability-motive-action-resistance-and-consequence-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model

or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if observed conformity proves acceptance or nonconformity proves knowledge.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `VDEM-DATA-001` - transport captured; captured bytes 122924; snapshot
sha256: cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- `UNODC-DATA-001` - transport captured; captured bytes 44411; snapshot
sha256: 35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- `OECD-DATA-EXPLORER-001` - transport captured; captured bytes 173987; snapshot
sha256: cdac198c1032dbb902cd9bfd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **compliance and resistance**, the consequence is practical and exact: Compliance and resistance remain distinct from ignorance, inability, concealment and coincidence. A scientific claim therefore has to expose the carrier `subject-rule-action-reason-carrier`, the relation `rule-to-behaviour-comparison`, the record `knowledge-capability-motive-action-resistance-and-consequence-held`, and the boundary `compliance-observation-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:d937c94fc7db7669dc87ad141c82e24bf169bcf5d6e24b02f4d00ee11055503b; independent certificate: sha256:7cad08491ab74f4c15c61894c89a84d98f7cf87a244ba27a461c59cf081156de; engine receipt: sha256:c9e98286efce43ca1d05239813f71d18955c9d9420d1071237c72d6422895580. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

24. Institutional change

Claim: SFT-SOCIAL-INSTITUTIONAL-CHANGE-001 **Family:** norm_institution_enforcement **Exact statement:** Institutional change is a source-dated transition retaining continuity, conflict and alternative accounts.

Forward chain and exact carrier

The generated carrier is `institution-prior-event-later-carrier`. Its required relation is `rule-practice-transition-relation`. The reconstruction must retain `versions-agents-events-continuities-conflicts-and-time-held`. Its declared empirical boundary is `historical-transition-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-ACTION-001, SFT-SOCIAL-COMPLIANCE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `institution-prior-event-later-carrier`.
- **boundary** - which population, place, period and method bound the statement: `historical-transition-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `rule-practice-transition-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `versions-agents-events-continuities-conflicts-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.

- **extension** - whether any fitted weight, exception or opaque oracle has been added: **no-extra-rule**.

Their ordered product gives the unique survivor:

`institution-prior-event-later-carrier__historical-transition-bounded__rule-practice-transition-relation__versions-agents-events-continuities-conflicts-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: **True**. Named-form uniqueness passed: **True**. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if a later state is used as its own causal history.

- `false_premise` - passed **True**: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed **True**: reject a changed registered source identity
- `tampered_artifact` - passed **True**: reject a missing, duplicate or additional survivor
- `boundary` - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **VDEM-DATA-001** - transport captured; captured bytes 122924; snapshot
sha256: cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- **UNODC-DATA-001** - transport captured; captured bytes 44411; snapshot
sha256: 35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- **OECD-DATA-EXPLORER-001** - transport captured; captured bytes 173987; snapshot
sha256: cdac198c1032dbb902cd9b9fd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **institutional change**, the consequence is practical and exact: Institutional change is a source-dated transition retaining continuity, conflict and alternative accounts. A scientific claim therefore has to expose the carrier

`institution-prior-event-later-carrier`, the relation `rule-practice-transition-relation`, the record `versions-agents-events-continuities-conflicts-and-time-held`, and the boundary

historical-transition-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:50951d82a43651cc20974ef942b18167fa0f5abf1885ef480a9047c9dad1354f; independent certificate: sha256:dc0906bfb8737219ab494d5aa7f0542bff6f847f64b3e08b7d8f882a4d5bcf8a; engine receipt: sha256:a3453592fdd413eeda7249449dec8f49cacb169203d83c7888ed55df399c3116. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

25. Trust relation

Claim: SFT-SOCIAL-TRUST-001 **Family:** trust_reputation_knowledge **Exact statement:** Trust is domain-specific reliance under vulnerability with history, action, outcome and breach retained.

Forward chain and exact carrier

The generated carrier is trustor-trustee-vulnerability-expectation-carrier. Its required relation is reliance-and-outcome-relation. The reconstruction must retain parties-domain-risk-history-action-and-breach-held. Its declared empirical boundary is trust-domain-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-INSTITUTIONAL-CHANGE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: trustor-trustee-vulnerability-expectation-carrier.
- **boundary** - which population, place, period and method bound the statement: trust-domain-bounded.
- **relation** - which generated interaction or comparison connects the carriers: reliance-and-outcome-relation.

- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
parties-domain-risk-history-action-and-breach-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

trustor-trustee-vulnerability-expectation-carrier__trust-domain-bounded__reliance-and-outcome-relation__parties-domain-risk-history-action-and-breach-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if familiarity, approval or one successful outcome proves general trust.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256: 32ceaf90269046a7bf55fdedb725d304909afac6e76dda8e1016a6d83f3898ba.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not

claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **trust relation**, the consequence is practical and exact: Trust is domain-specific reliance under vulnerability with history, action, outcome and breach retained. A scientific claim therefore has to expose the carrier **trustor-trustee-vulnerability-expectation-carrier**, the relation **reliance-and-outcome-relation**, the record **parties-domain-risk-history-action-and-breach-held**, and the boundary **trust-domain-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 80335b9b7fd6bd22545282ab797c6ff25b598e5539dcfb67ea4b37d8efe3b4c7; independent certificate: sha256: 7ba829a0a271c08771d63f39449d2bd15a55eccf6f6789397d3def4af322f0a5; engine receipt: sha256: 7d449223848c03df9221d66e6483af9123afd8bf01787c1c24e51500b5d9c9b3. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

26. Reputation

Claim: SFT-SOCIAL-REPUTATION-001 **Family:** trust_reputation_knowledge **Exact statement:** Reputation is a network of attributed reports whose sources, incentives, conflicts and audiences remain visible.

Forward chain and exact carrier

The generated carrier is **subject-observers-reports-carrier**. Its required relation is **report-aggregation-and-transmission-relation**. The reconstruction must retain **sources-audiences-content-time-conflict-and-incentives-held**. Its declared empirical boundary is **reputation-network-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-INSTITUTIONAL-CHANGE-001, SFT-SOCIAL-TRUST-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `subject-observers-reports-carrier`.
- **boundary** - which population, place, period and method bound the statement: `reputation-network-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `report-aggregation-and-transmission-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `sources-audiences-content-time-conflict-and-incentives-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`subject-observers-reports-carrier__reputation-network-bounded__report-aggregation-and-transmission-relation__sources-audiences-content-time-conflict-and-incentives-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if popularity or repeated rumor becomes verified character.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or

prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot sha256:32ceaf90269046a7bf55fdedb725d304909afac6e76dda8e1016a6d83f3898ba.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **reputation**, the consequence is practical and exact: Reputation is a network of attributed reports whose sources, incentives, conflicts and audiences remain visible. A scientific claim therefore has to expose the carrier subject-observers-reports-carrier, the relation report-aggregation-and-transmission-relation, the record sources-audiences-content-time-conflict-and-incentives-held, and the boundary reputation-network-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:1bf73d3007311d972eaf0455d23deb8b1d194f4c82d534794ddb0262445ed717; independent certificate: sha256:a8f73ab2857155fffb8ba76822bede1a00307ada8fce3235407f057279b4d0df; engine receipt: sha256:6fac43db587b30f5f66791a478d0c848b583c8dfefb18e4ed5b10e16daace7ef9. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

27. Testimony and credibility

Claim: SFT-SOCIAL-TESTIMONY-001 **Family:** trust_reputation_knowledge **Exact statement:** Testimony retains speaker access, claim, support, conflicts, audience and reception without credential substitution.

Forward chain and exact carrier

The generated carrier is speaker-claim-audience-evidence-carrier. Its required relation is testimony-to-belief-opportunity-relation. The reconstruction must retain identity-access-claim-support-conflict-and-reception-held. Its declared empirical boundary is testimony-event-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001,

SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-INSTITUTIONAL-CHANGE-001, SFT-SOCIAL-REPUTATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `speaker-claim-audience-evidence-carrier`.
- **boundary** - which population, place, period and method bound the statement: `testimony-event-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `testimony-to-belief-opportunity-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `identity-access-claim-support-conflict-and-reception-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`speaker-claim-audience-evidence-carrier__testimony-event-bounded__testimony-to-belief-opportunity-relation__identity-access-claim-support-conflict-and-reception-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if status alone establishes truth or marginal status establishes falsity.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor

- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256: 32ceaf90269046a7bf55fdedb725d304909afac6e76dda8e1016a6d83f3898ba.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **testimony and credibility**, the consequence is practical and exact: Testimony retains speaker access, claim, support, conflicts, audience and reception without credential substitution. A scientific claim therefore has to expose the carrier `speaker-claim-audience-evidence-carrier`, the relation `testimony-to-belief-opportunity-relation`, the record `identity-access-claim-support-conflict-and-reception-held`, and the boundary `testimony-event-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256: b6ac1ca14619112a72e8d95677ac9729fd78fa339ca4442ed334ffa727a1b4d1; independent
certificate: sha256: 22b88834c029aeea55d3e72926f18797abc6543e64db7aefdbc18754c0998942; engine
receipt: sha256: 4b52d15532ed596150734988a9cfd8ce768f6190152d912d5420a4fb4468c0b5. The
independent implementation recomputed the candidate product: **True**. All external rows were preserved: **True**.

28. Collective knowledge

Claim: SFT-SOCIAL-COLLECTIVE-KNOWLEDGE-001 **Family:** `trust_reputation_knowledge` **Exact statement:**
Collective knowledge is distributed, recorded and validated content with contribution, access and disagreement retained.

Forward chain and exact carrier

The generated carrier is `distributed-agents-records-validation-carrier`. Its required relation is `composition-and-retrieval-relation`. The reconstruction must retain `contributors-methods-records-disagreements-access-and-time-held`. Its declared empirical boundary is `knowledge-system-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation

that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-INSTITUTIONAL-CHANGE-001, SFT-SOCIAL-TESTIMONY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: distributed-agents-records-validation-carrier.
- **boundary** - which population, place, period and method bound the statement: knowledge-system-bounded.
- **relation** - which generated interaction or comparison connects the carriers: composition-and-retrieval-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: contributors-methods-records-disagreements-access-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

distributed-agents-records-validation-carrier__knowledge-system-bounded__composition-and-retrieval-relation__contributors-methods-records-disagreements-access-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if consensus vote replaces evidence or inaccessible content is called shared knowledge.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256: 32ceaf90269046a7bf55fdebd725d304909afac6e76dda8e1016a6d83f3898ba.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective knowledge**, the consequence is practical and exact: Collective knowledge is distributed, recorded and validated content with contribution, access and disagreement retained. A scientific claim therefore has to expose the carrier `distributed-agents-records-validation-carrier`, the relation `composition-and-retrieval-relation`, the record `contributors-methods-records-disagreements-access-and-time-held`, and the boundary `knowledge-system-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: d190b023d81a3477be7b1c0f1112755ba3a2f0f9743b9549f8465e2b2d9cb92a; independent certificate: sha256: 4a542bc0e9b7626f505f2cb1da3d2550d48034a46bd4499c6e340fdd6848a6a2; engine receipt: sha256: fcefd23cfc7590e92f561b84763b94bc6b65e5b19c6efa5a317cd329c2c4af1. The

independent implementation recomputed the candidate product: **True**. All external rows were preserved: **True**.

29. Expertise evidence boundary

Claim: SFT-SOCIAL-EXPERTISE-001 **Family:** trust_reputation_knowledge **Exact statement:** Expertise is domain- and task-bound competence evidenced by methods and performance, not a universal authority token.

Forward chain and exact carrier

The generated carrier is **agent-domain-performance-record-carrier**. Its required relation is **competence-to-task-relation**. The reconstruction must retain **domain-training-method-performance-errors-and-updates-held**. Its declared empirical boundary is **expertise-domain-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-INSTITUTIONAL-CHANGE-001, SFT-SOCIAL-COLLECTIVE-KNOWLEDGE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: **agent-domain-performance-record-carrier**.
- **boundary** - which population, place, period and method bound the statement: **expertise-domain-bounded**.
- **relation** - which generated interaction or comparison connects the carriers: **competence-to-task-relation**.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: **domain-training-method-performance-errors-and-updates-held**.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: **evidence-and-normative-class-explicit**.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: **root-bound-forward-forcing**.
- **generality** - whether finite extension preserves every context rather than universalising one case: **positive-finite-extension-retains-all-contexts**.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: **no-extra-rule**.

Their ordered product gives the unique survivor:

agent-domain-performance-record-carrier__expertise-domain-bounded__competence-to-task-relation__domain-training-method-performance-errors-and-updates-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject credential, title or confidence as sufficient evidence of a claim.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256: 32ceaf90269046a7bf55fdebd725d304909afac6e76dda8e1016a6d83f3898ba.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **expertise evidence boundary**, the consequence is practical and exact: Expertise is domain- and task-bound competence evidenced by methods and performance, not a universal authority token. A scientific claim therefore has to expose the carrier `agent-domain-performance-record-carrier`, the relation `competence-to-task-relation`, the record `domain-training-method-performance-errors-and-updates-held`, and the boundary `expertise-domain-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:5426f809ebf39d48907c25cf67c3720b3e9719d4cd9d901486c427d3ee2867f1; independent certificate: sha256:c35c9ba8faeea62b5239f206a61d309b182eb4ef56c97cd994a761a2dd00619a; engine receipt: sha256:bbde304ac7029a5c12d6256e624245f3fbfc8aeb6e6629f72843188ac0541ee0. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

30. False and unsupported social information

Claim: SFT-SOCIAL-MISINFORMATION-001 **Family:** trust_reputation_knowledge **Exact statement:** False, unsupported, disputed and corrected claims remain distinct and source-dated.

Forward chain and exact carrier

The generated carrier is `claim-source-channel-evidence-carrier`. Its required relation is `claim-to-support-comparison`. The reconstruction must retain `versions-sources-support-corrections-reach-and-harm-held`. Its declared empirical boundary is `claim-and-time-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-INSTITUTIONAL-CHANGE-001, SFT-SOCIAL-EXPERTISE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `claim-source-channel-evidence-carrier`.
- **boundary** - which population, place, period and method bound the statement: `claim-and-time-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `claim-to-support-comparison`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `versions-sources-support-corrections-reach-and-harm-held`.

- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
claim-source-channel-evidence-carrier__claim-and-time-bounded__claim-to-support-compari
son__versions-sources-support-corrections-reach-and-harm-held__evidence-and-normative-c
lass-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contex
ts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if dissent is labelled false by status or if later correction erases prior reach.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `GSS-NORC-001` - transport `captured`; captured bytes 67309; snapshot `sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1`.
- `ICPSR-DATA-001` - transport `failed_preserved`; captured bytes 0; snapshot `unresolved and preserved`.
- `CROSSREF-RECORDS-001` - transport `captured`; captured bytes 69898; snapshot `sha256:32ceaf90269046a7bf55fdebd725d304909afac6e76dda8e1016a6d83f3898ba`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and

adverse rows retained.

Scientific meaning

For **false and unsupported social information**, the consequence is practical and exact: False, unsupported, disputed and corrected claims remain distinct and source-dated. A scientific claim therefore has to expose the carrier `claim-source-channel-evidence-carrier`, the relation `claim-to-support-comparison`, the record `versions-sources-support-corrections-reach-and-harm-held`, and the boundary `claim-and-time-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

`sha256:33987505914509e27797b58608a0edae897eb409b1ba1062a7e1e74df0816580`; independent certificate: `sha256:16004d8efe3b1bdc1b4cdd00ce52f70a10206fc37b64465facdb303488b350e4`; engine receipt: `sha256:8b2d26d118ad33a5180de1fa62f95ba04009100be7cf9f438d2e7f03f3b83f51`. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

31. Exchange

Claim: SFT-SOCIAL-EXCHANGE-001 **Family:** `exchange_allocation_distribution` **Exact statement:** Exchange retains parties, items, rights, terms, timing, costs and failed transfers.

Forward chain and exact carrier

The generated carrier is `parties-items-rights-transfer-carrier`. Its required relation is `reciprocal-or-directed-transfer-relation`. The reconstruction must retain `parties-items-rights-terms-time-costs-and-failure-held`. Its declared empirical boundary is `exchange-event-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MISINFORMATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `parties-items-rights-transfer-carrier`.
- **boundary** - which population, place, period and method bound the statement: `exchange-event-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `reciprocal-or-directed-transfer-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `parties-items-rights-terms-time-costs-and-failure-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`parties-items-rights-transfer-carrier__exchange-event-bounded__reciprocal-or-directed-transfer-relation__parties-items-rights-terms-time-costs-and-failure-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if a price alone describes rights, coercion, quality or distribution.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- **WORLD-BANK-WDI-001** - transport captured; captured bytes 1874098; snapshot sha256: 4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- **ILOSTAT-001** - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- **UNSD-DEMOGRAPHIC-001** - transport captured; captured bytes 88501; snapshot sha256: cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **exchange**, the consequence is practical and exact: Exchange retains parties, items, rights, terms, timing, costs and failed transfers. A scientific claim therefore has to expose the carrier **parties-items-rights-transfer-carrier**, the relation **reciprocal-or-directed-transfer-relation**, the record **parties-items-rights-terms-time-costs-and-failure-held**, and the boundary **exchange-event-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: d8e8b8ca8aa35447912697c936ca7dbe3ff3272094698b0138b4d77d48318da7; independent certificate: sha256: 6033631925cfe63cc88fd9857d98eb4a8225dbaf56d306653a56f6bf42479452; engine receipt: sha256: df46ce2287ea5a75408e543fe7a36fd714b95f8fd11c611b97a3d1875c600a58. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

32. Social valuation

Claim: SFT-SOCIAL-VALUE-001 **Family:** exchange_allocation_distribution **Exact statement:** Valuation is agent-, purpose-, option- and context-bound and cannot become an intrinsic universal scalar by declaration.

Forward chain and exact carrier

The generated carrier is **agent-object-purpose-context-carrier**. Its required relation is **preference-and-tradeoff-relation**. The reconstruction must retain **agent-purpose-options-information-time-and-conflict-held**. Its declared empirical boundary is **valuation-context-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001,

SFT-SOCIAL-MISINFORMATION-001, SFT-SOCIAL-EXCHANGE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agent-object-purpose-context-carrier`.
- **boundary** - which population, place, period and method bound the statement: `valuation-context-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `preference-and-tradeoff-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `agent-purpose-options-information-time-and-conflict-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`agent-object-purpose-context-carrier__valuation-context-bounded__preference-and-tradeoff-relation__agent-purpose-options-information-time-and-conflict-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if market price, vote or survey mean is equated with universal human value.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from

the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `WORLD-BANK-WDI-001` - transport captured; captured bytes `1874098`; snapshot `sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927`.
- `ILOSTAT-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved` and `preserved`.
- `UNSD-DEMOGRAPHIC-001` - transport captured; captured bytes `88501`; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social valuation**, the consequence is practical and exact: Valuation is agent-, purpose-, option- and context-bound and cannot become an intrinsic universal scalar by declaration. A scientific claim therefore has to expose the carrier `agent-object-purpose-context-carrier`, the relation `preference-and-tradeoff-relation`, the record `agent-purpose-options-information-time-and-conflict-held`, and the boundary `valuation-context-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: `sha256:47babbb5d3db640af04b574c00be5800d752315be65e451c840542df19e2cfe1`; independent certificate: `sha256:24257056e9390a8c5a061a3c61f9e3650a97b7029afe7f7be7615e7402aff36d`; engine receipt: `sha256:122e9236d3b52979be9fe329ea942a6fd431eb52b6a405859ce2bd5cf36b5a84`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

33. Resource allocation

Claim: `SFT-SOCIAL-ALLOCATION-001` **Family:** `exchange_allocation_distribution` **Exact statement:** Allocation retains total resources, claimants, rule, priority, outcome, unmet claims and losses.

Forward chain and exact carrier

The generated carrier is `resources-claimants-rule-outcome-carrier`. Its required relation is `rule-to-distribution-relation`. The reconstruction must retain `stocks-claims-rule-priority-outcomes-losses-and-time-held`. Its declared empirical boundary is `allocation-boundary-declared`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`,

SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MISINFORMATION-001, SFT-SOCIAL-VALUE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `resources-claimants-rule-outcome-carrier`.
- **boundary** - which population, place, period and method bound the statement: `allocation-boundary-declared`.
- **relation** - which generated interaction or comparison connects the carriers: `rule-to-distribution-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `stocks-claims-rule-priority-outcomes-losses-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`resources-claimants-rule-outcome-carrier__allocation-boundary-declared__rule-to-distribution-relation__stocks-claims-rule-priority-outcomes-losses-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if distributed shares omit the stock, denominator or excluded claimants.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `WORLD-BANK-WDI-001` - transport captured; captured bytes 1874098; snapshot `sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927`.
- `ILOSTAT-001` - transport `failed_preserved`; captured bytes 0; snapshot `unresolved` and `preserved`.
- `UNSD-DEMOGRAPHIC-001` - transport captured; captured bytes 88501; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **resource allocation**, the consequence is practical and exact: Allocation retains total resources, claimants, rule, priority, outcome, unmet claims and losses. A scientific claim therefore has to expose the carrier `resources-claimants-rule-outcome-carrier`, the relation `rule-to-distribution-relation`, the record `stocks-claims-rule-priority-outcomes-losses-and-time-held`, and the boundary `allocation-boundary-declared` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

`sha256:d9ddcf56ef58df852c3b6b7836d0974cd0f40e695208ffc4702afd771e7f0519`; independent certificate: `sha256:edd12741785091b149d384f75d5aec2da2a801cfff090c8128a248a3670154539`; engine receipt: `sha256:92cc1799ee389a64befd0ba46bc3ff916ead60a0b03f32aecafd9bef4a4c9136`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

34. Collective production

Claim: `SFT-SOCIAL-PRODUCTION-001` **Family:** `exchange_allocation_distribution` **Exact statement:** Production retains labour, material, knowledge, ownership, transformation, output and loss.

Forward chain and exact carrier

The generated carrier is **contributors-input-process-output-carrier**. Its required relation is **input-transformation-output-relation**. The reconstruction must retain **labour-material-knowledge-ownership-losses-and-time-held**. Its declared empirical boundary is **production-process-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MISINFORMATION-001, SFT-SOCIAL-ALLOCATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: **contributors-input-process-output-carrier**.
- **boundary** - which population, place, period and method bound the statement: **production-process-bounded**.
- **relation** - which generated interaction or comparison connects the carriers: **input-transformation-output-relation**.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: **labour-material-knowledge-ownership-losses-and-time-held**.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: **evidence-and-normative-class-explicit**.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: **root-bound-forward-forcing**.
- **generality** - whether finite extension preserves every context rather than universalising one case: **positive-finite-extension-retains-all-contexts**.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: **no-extra-rule**.

Their ordered product gives the unique survivor:

contributors-input-process-output-carrier__production-process-bounded__input-transformation-output-relation__labour-material-knowledge-ownership-losses-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the

result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if output alone erases contributors, unpaid inputs, waste or ownership.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `WORLD-BANK-WDI-001` - transport captured; captured bytes 1874098; snapshot `sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927`.
- `ILOSTAT-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- `UNSD-DEMOGRAPHIC-001` - transport captured; captured bytes 88501; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective production**, the consequence is practical and exact: Production retains labour, material, knowledge, ownership, transformation, output and loss. A scientific claim therefore has to expose the carrier `contributors-input-process-output-carrier`, the relation `input-transformation-output-relation`, the record `labour-material-knowledge-ownership-losses-and-time-held`, and the boundary `production-process-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: df217c3d6bf052768a084dc7f54301bd2783ad6b9c1c58b7721c1b4e55047102; independent certificate: sha256: fe2865499c449433ede52033b755bfc29b7634789992257111c20d2d6324e59; engine receipt: sha256: ae53c6cd5ece19bc49289d53a331d7283bedc3c36f6b065fe93cdf63a1df41a7. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

35. Distribution and inequality

Claim: SFT-SOCIAL-DISTRIBUTION-001 **Family:** exchange_allocation_distribution **Exact statement:** Distribution retains population, total stock, shares, access, time and missing members.

Forward chain and exact carrier

The generated carrier is `population-resource-share-carrier`. Its required relation is `member-to-share-relation`. The reconstruction must retain `population-frame-stock-shares-access-time-and-missing-held`. Its declared empirical boundary is `distribution-population-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MISINFORMATION-001, SFT-SOCIAL-PRODUCTION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `population-resource-share-carrier`.
- **boundary** - which population, place, period and method bound the statement: `distribution-population-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `member-to-share-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `population-frame-stock-shares-access-time-and-missing-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.

- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
population-resource-share-carrier__distribution-population-bounded__member-to-share-relation__population-frame-stock-shares-access-time-and-missing-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if an average, rank or index replaces the complete share distribution.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `WORLD-BANK-WDI-001` - transport captured; captured bytes 1874098; snapshot `sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927`.
- `ILOSTAT-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- `UNSD-DEMOGRAPHIC-001` - transport captured; captured bytes 88501; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **distribution and inequality**, the consequence is practical and exact: Distribution retains population, total stock, shares, access, time and missing members. A scientific claim therefore has to expose the carrier `population-resource-share-carrier`, the relation `member-to-share-relation`, the record `population-frame-stock-shares-access-time-and-missing-held`, and the boundary

distribution-population-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:d0ad85eaa66d900468c7e7b34afa33159deb383dcfe5488c216521dbea44ee01; independent certificate: sha256:6156cda5d927062ab0eb76ec9898486c89c3d6c4fbbd5a8a6f3093f1bebc8c; engine receipt: sha256:94cd29960f46745e6f2ac0a7dc569aeef6d1918e36572172dfc28005de2bd6a. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

36. Social mobility

Claim: SFT-SOCIAL-MOBILITY-001 **Family:** exchange_allocation_distribution **Exact statement:** Mobility is a cohort- and measure-bound transition retaining paths, attrition and opportunity structure.

Forward chain and exact carrier

The generated carrier is **agent-origin-destination-time-carrier**. Its required relation is **state-transition-and-opportunity-relation**. The reconstruction must retain **cohort-frame-origin-destination-path-attrition-and-time-held**. Its declared empirical boundary is **cohort-and-measure-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MISINFORMATION-001, SFT-SOCIAL-DISTRIBUTION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: **agent-origin-destination-time-carrier**.
- **boundary** - which population, place, period and method bound the statement: **cohort-and-measure-bounded**.
- **relation** - which generated interaction or comparison connects the carriers: **state-transition-and-opportunity-relation**.

- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
cohort-frame-origin-destination-path-attrition-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

agent-origin-destination-time-carrier__cohort-and-measure-bounded__state-transition-and
-opportunity-relation__cohort-frame-origin-destination-path-attrition-and-time-held__ev
idence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extens
ion-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject cross-sectional difference as individual mobility or omit those lost to follow-up.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `WORLD-BANK-WDI-001` - transport captured; captured bytes 1874098; snapshot
sha256: 4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- `ILOSTAT-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- `UNSD-DEMOGRAPHIC-001` - transport captured; captured bytes 88501; snapshot
sha256: cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not

claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social mobility**, the consequence is practical and exact: Mobility is a cohort- and measure-bound transition retaining paths, attrition and opportunity structure. A scientific claim therefore has to expose the carrier **agent-origin-destination-time-carrier**, the relation **state-transition-and-opportunity-relation**, the record **cohort-frame-origin-destination-path-attrition-and-time-held**, and the boundary **cohort-and-measure-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:27674d075ad945e9ed09dcfcfd45533672fcf7379dec377c005f18a96b70d9db; independent certificate: sha256:1089d0b29d4263984680d667e9ab267ad414cd77e8205fed70afe5dd04ee1205; engine receipt: sha256:c5ed76c9b77ad263fb8bd1c66d620aa54b73c373e4fdbb0aa7111d912b219823. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

37. Power relation

Claim: SFT-SOCIAL-POWER-001 **Family:** power_authority_exclusion **Exact statement:** Power is a directional capacity to alter another's available actions, retaining resources, dependency and resistance.

Forward chain and exact carrier

The generated carrier is **actor-target-capability-dependency-carrier**. Its required relation is **capacity-to-shape-options-relation**. The reconstruction must retain **actors-resources-options-dependencies-actions-and-resistance-held**. Its declared empirical boundary is **power-domain-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: actor-target-capability-dependency-carrier.
- **boundary** - which population, place, period and method bound the statement: power-domain-bounded.
- **relation** - which generated interaction or comparison connects the carriers: capacity-to-shape-options-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: actors-resources-options-dependencies-actions-and-resistance-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

actor-target-capability-dependency-carrier__power-domain-bounded__capacity-to-shape-options-relation__actors-resources-options-dependencies-actions-and-resistance-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is **depth_independent**. Minimality passed: **True**. Named-form uniqueness passed: **True**. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject status, wealth or outcome alone as complete power evidence.

- **false_premise** - passed **True**: reject the generated form lacking the complete physical carrier
- **tampered_source** - passed **True**: reject a changed registered source identity
- **tampered_artifact** - passed **True**: reject a missing, duplicate or additional survivor
- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: **source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence**. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNHCR-REFUGEE-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **power relation**, the consequence is practical and exact: Power is a directional capacity to alter another's available actions, retaining resources, dependency and resistance. A scientific claim therefore has to expose the carrier **actor-target-capability-dependency-carrier**, the relation **capacity-to-shape-options-relation**, the record **actors-resources-options-dependencies-actions-and-resistance-held**, and the boundary **power-domain-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:d42e21287c383b6f1aca83483562065d5ba804fb29d1b752f4adab6714c06494; independent certificate: sha256:05a66bce16dd11210d1184b14d825fe09bd3e18a1ff3e7dacc1bfd7da1a7b2be; engine receipt: sha256:4c0a80d595e1a96fcd44f40e8970537a863559f2c0af12e7d1f52726586349f. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

38. Authority

Claim: SFT-SOCIAL-AUTHORITY-001 **Family:** power_authority_exclusion **Exact statement:** Authority is scoped decision capacity with source, procedure, recognition and contestation retained.

Forward chain and exact carrier

The generated carrier is **office-subject-rule-recognition-carrier**. Its required relation is **authorized-decision-relation**. The reconstruction must retain **source-scope-procedure-recognition-contestation-and-time-held**. Its declared empirical boundary is **authority-jurisdiction-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001,

SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001, SFT-SOCIAL-POWER-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `office-subject-rule-recognition-carrier`.
- **boundary** - which population, place, period and method bound the statement: `authority-jurisdiction-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `authorized-decision-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `source-scope-procedure-recognition-contestation-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`office-subject-rule-recognition-carrier__authority-jurisdiction-bounded__authorized-decision-relation__source-scope-procedure-recognition-contestation-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject effective coercion as legitimate authority or formal title as effective control.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from

the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot sha256: cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNHCR-REFUGEE-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot sha256: cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **authority**, the consequence is practical and exact: Authority is scoped decision capacity with source, procedure, recognition and contestation retained. A scientific claim therefore has to expose the carrier `office-subject-rule-recognition-carrier`, the relation `authorized-decision-relation`, the record `source-scope-procedure-recognition-contestation-and-time-held`, and the boundary `authority-jurisdiction-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256: 9f62e64a12d9de1be7460efde638a385261299869188537669c046435db3c6dd; independent certificate: sha256: 6d8a0f5be84561352ab211fb21676430c19dc5f33d2d746095e965806432cd74; engine receipt: sha256: 0ef12b8ddff302feeee5b78871251d4fc32f15a1a4339cbb8e6276356ffb202c. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

39. Social dependency

Claim: SFT-SOCIAL-DEPENDENCY-001 **Family:** `power_authority_exclusion` **Exact statement:** Dependency retains the needed resource, controller, alternatives, costs and failures.

Forward chain and exact carrier

The generated carrier is `agents-needed-resource-alternatives-carrier`. Its required relation is `access-and-substitution-relation`. The reconstruction must retain `resource-controller-alternatives-costs-time-and-failures-held`. Its declared empirical boundary is `dependency-resource-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,

SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001,
 SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001,
 SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001,
 SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001,
 SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001,
 SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001,
 SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001,
 SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001,
 SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001,
 SFT-SOCIAL-AUTHORITY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-needed-resource-alternatives-carrier`.
- **boundary** - which population, place, period and method bound the statement: `dependency-resource-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `access-and-substitution-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `resource-controller-alternatives-costs-time-and-failures-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`agents-needed-resource-alternatives-carrier__dependency-resource-bounded__access-and-substitution-relation__resource-controller-alternatives-costs-time-and-failures-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject repeated interaction or correlation as dependency without the alternative boundary.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `VDEM-DATA-001` - transport captured; captured bytes 122924; snapshot `sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c`.
- `UNHCR-REFUGEE-DATA-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- `UNSD-DEMOGRAPHIC-001` - transport captured; captured bytes 88501; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social dependency**, the consequence is practical and exact: Dependency retains the needed resource, controller, alternatives, costs and failures. A scientific claim therefore has to expose the carrier `agents-needed-resource-alternatives-carrier`, the relation `access-and-substitution-relation`, the record `resource-controller-alternatives-costs-time-and-failures-held`, and the boundary `dependency-resource-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

`sha256:03ad978cfa831c6fb7b29b76aa1971b2feb47e566d561f15e3559f99976534d9`; independent certificate: `sha256:68bf2531feb501fc844780174cab3d40ee9bb234fa4dfa303af3b06a50a69650`; engine receipt: `sha256:c1c1151da996efecdc23edb781a9fbaf4077c38bc76b903d61c99218f53f2163`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

40. Exclusion

Claim: `SFT-SOCIAL-EXCLUSION-001` **Family:** `power_authority_exclusion` **Exact statement:** Exclusion retains opportunity, eligibility, gate, decision, appeal and unobserved discouraged cases.

Forward chain and exact carrier

The generated carrier is **agent-opportunity-gate-decision-carrier**. Its required relation is **eligibility-access-and-denial-relation**. The reconstruction must retain **criteria-gatekeeper-process-outcome-appeal-and-hidden-cases-held**. Its declared empirical boundary is **access-system-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001, SFT-SOCIAL-DEPENDENCY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: **agent-opportunity-gate-decision-carrier**.
- **boundary** - which population, place, period and method bound the statement: **access-system-bounded**.
- **relation** - which generated interaction or comparison connects the carriers: **eligibility-access-and-denial-relation**.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: **criteria-gatekeeper-process-outcome-appeal-and-hidden-cases-held**.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: **evidence-and-normative-class-explicit**.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: **root-bound-forward-forcing**.
- **generality** - whether finite extension preserves every context rather than universalising one case: **positive-finite-extension-retains-all-contexts**.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: **no-extra-rule**.

Their ordered product gives the unique survivor:

agent-opportunity-gate-decision-carrier__access-system-bounded__eligibility-access-and-denial-relation__criteria-gatekeeper-process-outcome-appeal-and-hidden-cases-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the

result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject if only successful applicants define access or if absence from a system proves free choice.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `VDEM-DATA-001` - transport `captured`; captured bytes 122924; snapshot `sha256:cae9debd85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c`.
- `UNHCR-REFUGEE-DATA-001` - transport `failed_preserved`; captured bytes 0; snapshot `unresolved` and `preserved`.
- `UNSD-DEMOGRAPHIC-001` - transport `captured`; captured bytes 88501; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **exclusion**, the consequence is practical and exact: Exclusion retains opportunity, eligibility, gate, decision, appeal and unobserved discouraged cases. A scientific claim therefore has to expose the carrier `agent-opportunity-gate-decision-carrier`, the relation `eligibility-access-and-denial-relation`, the record `criteria-gatekeeper-process-outcome-appeal-and-hidden-cases-held`, and the boundary `access-system-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:02341d7fa6241cee664c3be50f2ed4dbad68b9fcbbef88a39704a55442e22bc3; independent certificate: sha256:969ca761928a47b3d651667c6330dad3ddfd8c8dc0bac1cf09fd906e2e12acd5; engine receipt: sha256:a706086b718bc0c37342a9586801a6b6e53528d197bbd7e592a47249c98d23b4. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

41. Stratification

Claim: SFT-SOCIAL-STRATIFICATION-001 **Family:** power_authority_exclusion **Exact statement:** Stratification is persistent ordered access across a declared population and resource set with mobility retained.

Forward chain and exact carrier

The generated carrier is `population-positions-resources-relations-carrier`. Its required relation is `ordered-access-and-reproduction-relation`. The reconstruction must retain `frame-groups-resources-mobility-institutions-and-time-held`. Its declared empirical boundary is `stratification-context-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001, SFT-SOCIAL-EXCLUSION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `population-positions-resources-relations-carrier`.
- **boundary** - which population, place, period and method bound the statement: `stratification-context-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `ordered-access-and-reproduction-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `frame-groups-resources-mobility-institutions-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.

- **extension** - whether any fitted weight, exception or opaque oracle has been added: **no-extra-rule**.

Their ordered product gives the unique survivor:

population-positions-resources-relations-carrier__stratification-context-bounded__ordered-access-and-reproduction-relation__frame-groups-resources-mobility-institutions-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is **depth_independent**. Minimality passed: **True**. Named-form uniqueness passed: **True**. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject universal hierarchy from one measure, institution or historical context.

- **false_premise** - passed **True**: reject the generated form lacking the complete physical carrier
- **tampered_source** - passed **True**: reject a changed registered source identity
- **tampered_artifact** - passed **True**: reject a missing, duplicate or additional survivor
- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: **source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence**. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **VDEM-DATA-001** - transport **captured**; captured bytes 122924; snapshot
sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- **UNHCR-REFUGEE-DATA-001** - transport **failed_preserved**; captured bytes 0; snapshot **unresolved** and **preserved**.
- **UNSD-DEMOGRAPHIC-001** - transport **captured**; captured bytes 88501; snapshot
sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **stratification**, the consequence is practical and exact: Stratification is persistent ordered access across a declared population and resource set with mobility retained. A scientific claim therefore has to expose the carrier **population-positions-resources-relations-carrier**, the relation **ordered-access-and-reproduction-relation**, the record

frame-groups-resources-mobility-institutions-and-time-held, and the boundary stratification-context-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:0bc297cd2916ec76cd06445d82a2034411a22a4c102f72082a2cbbbe15f14aa16; independent certificate: sha256:1d3743391415860558b3ba7e53ce11d62175096f48a1f95ed3ac42be127b07c7; engine receipt: sha256:bdace614acbf29d3c8007db7999b6fdfe0cf4079dd7c93d4785cc2aa5127a5da. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

42. Scientific and institutional gatekeeping

Claim: SFT-SOCIAL-GATEKEEPING-001 **Family:** power_authority_exclusion **Exact statement:** Gatekeeping evidence retains criteria, costs, credentials, decisions, appeals and contributions made invisible by exclusion.

Forward chain and exact carrier

The generated carrier is contributor-gate-resource-decision-carrier. Its required relation is access-selection-and-recognition-relation. The reconstruction must retain criteria-costs-credentials-decisions-appeals-and-lost-contributions-held. Its declared empirical boundary is institution-and-period-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001, SFT-SOCIAL-STRATIFICATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: contributor-gate-resource-decision-carrier.
- **boundary** - which population, place, period and method bound the statement: institution-and-period-bounded.

- **relation** - which generated interaction or comparison connects the carriers:
access-selection-and-recognition-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
criteria-costs-credentials-decisions-appeals-and-lost-contributions-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

contributor-gate-resource-decision-carrier__institution-and-period-bounded__access-selection-and-recognition-relation__criteria-costs-credentials-decisions-appeals-and-lost-contributions-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject prestige as scientific evidence or criticism of prestige as automatic admission of a claim.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `VDEM-DATA-001` - transport captured; captured bytes 122924; snapshot `sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c`.
- `UNHCR-REFUGEE-DATA-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

- **UNSD-DEMOGRAPHIC-001** - transport captured; captured bytes 88501; snapshot
sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **scientific and institutional gatekeeping**, the consequence is practical and exact: Gatekeeping evidence retains criteria, costs, credentials, decisions, appeals and contributions made invisible by exclusion. A scientific claim therefore has to expose the carrier **contributor-gate-resource-decision-carrier**, the relation **access-selection-and-recognition-relation**, the record **criteria-costs-credentials-decisions-appeals-and-lost-contributions-held**, and the boundary **institution-and-period-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:7d165494602d39b9e70ad90e6115c4ce8f177a3abab13ed4f1af1eb1128266ac; independent certificate: sha256:32e7f28723577ddba3fb1c574fd2b654e273181f9c6e73ad16b2d0dbbc00e362; engine receipt: sha256:9d1f3fb1cc3eb7127d43c77fd7f279c9f22c1c8f02e50075e4aae47f0589b4c2. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

43. Social network

Claim: SFT-SOCIAL-SOCIAL-NETWORK-001 **Family:** network_diffusion_transition **Exact statement:** A social network retains node identity, link meaning, direction, time, sampling and missing links.

Forward chain and exact carrier

The generated carrier is **agents-typed-links-time-carrier**. Its required relation is **graph-and-interaction-relation**. The reconstruction must retain **nodes-link-types-direction-weights-time-and-missing-held**. Its declared empirical boundary is **network-construction-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GATEKEEPING-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-typed-links-time-carrier`.
- **boundary** - which population, place, period and method bound the statement: `network-construction-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `graph-and-interaction-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `nodes-link-types-direction-weights-time-and-missing-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`agents-typed-links-time-carrier__network-construction-bounded__graph-and-interaction-relation__nodes-link-types-direction-weights-time-and-missing-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject an abstract edge as a social tie without measurement or infer absent links outside the sample.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social network**, the consequence is practical and exact: A social network retains node identity, link meaning, direction, time, sampling and missing links. A scientific claim therefore has to expose the carrier `agents-typed-links-time-carrier`, the relation `graph-and-interaction-relation`, the record `nodes-link-types-direction-weights-time-and-missing-held`, and the boundary `network-construction-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256:31009db1a0f6ef2e5446e5a37ce6c1f8fc68e72f194e00817b46d23364e4c963; independent
certificate: sha256:4a76d92075babced03efcfe72fedfe449ca485be46ce90f02ff6a29a3e9e7d77; engine
receipt: sha256:1d731ec9187e36749fd901e73d419bf40d6f0c5757d5b5db3d97b44eab87c5bc. The
independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

44. Social diffusion

Claim: SFT-SOCIAL-DIFFUSION-001 **Family:** `network_diffusion_transition` **Exact statement:** Diffusion retains source, path, exposure opportunity, adoption, nonadoption and external inputs.

Forward chain and exact carrier

The generated carrier is `content-agents-links-adoptions-carrier`. Its required relation is `exposure-transmission-adoption-relation`. The reconstruction must retain `source-path-opportunity-time-nonadoption-and-external-input-held`. Its declared empirical boundary is `diffusion-process-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001,

SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GATEKEEPING-001, SFT-SOCIAL-SOCIAL-NETWORK-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `content-agents-links-adoptions-carrier`.
- **boundary** - which population, place, period and method bound the statement: `diffusion-process-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `exposure-transmission-adoption-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `source-path-opportunity-time-nonadoption-and-external-input-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`content-agents-links-adoptions-carrier__diffusion-process-bounded__exposure-transmission-adoption-relation__source-path-opportunity-time-nonadoption-and-external-input-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject temporal clustering as peer transmission or exposure as adoption.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier

- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot `sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1`.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot `sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social diffusion**, the consequence is practical and exact: Diffusion retains source, path, exposure opportunity, adoption, nonadoption and external inputs. A scientific claim therefore has to expose the carrier `content-agents-links-adoptions-carrier`, the relation `exposure-transmission-adoption-relation`, the record `source-path-opportunity-time-nonadoption-and-external-input-held`, and the boundary `diffusion-process-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: `sha256:c1c60d5b76ac2b904a34d662bc966cd5f3d0840c70b5793b33b073e036677666`; independent certificate: `sha256:915a342c552e10508f6482be5901b42ec4249ea04eed2bf04bf476164928f666`; engine receipt: `sha256:67f740214a3cb3f17d5980370451e1a0229775652bfb7552b53a50c04fd9708b`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

45. Collective synchronization

Claim: SFT-SOCIAL-SYNCHRONIZATION-001 **Family:** `network_diffusion_transition` **Exact statement:** Synchronization is recurrent phase relation with coupling, lag, breaks and common-cause controls retained.

Forward chain and exact carrier

The generated carrier is `agents-rhythms-coupling-carrier`. Its required relation is `phase-recurrence-and-coupling-relation`. The reconstruction must retain

states-times-links-lags-breaks-and-controls-held. Its declared empirical boundary is synchrony-window-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GATEKEEPING-001, SFT-SOCIAL-DIFFUSION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: agents-rhythms-coupling-carrier.
- **boundary** - which population, place, period and method bound the statement: synchrony-window-bounded.
- **relation** - which generated interaction or comparison connects the carriers: phase-recurrence-and-coupling-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: states-times-links-lags-breaks-and-controls-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

agents-rhythms-coupling-carrier__synchrony-window-bounded__phase-recurrence-and-coupling-relation__states-times-links-lags-breaks-and-controls-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case

universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject coincident peaks or shared schedules as endogenous synchronization.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256 : f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256 : 5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective synchronization**, the consequence is practical and exact: Synchronization is recurrent phase relation with coupling, lag, breaks and common-cause controls retained. A scientific claim therefore has to expose the carrier `agents-rhythms-coupling-carrier`, the relation `phase-recurrence-and-coupling-relation`, the record `states-times-links-lags-breaks-and-controls-held`, and the boundary `synchrony-window-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256 : cb4ac5c069322e98e98291bb2fee7ba9f9695b6fd4aa44ca872fa1c56643b3ad; independent

certificate: sha256 : f93206d0491dbd6dbc5bec41de066242248008603d5a3ff5e408590a68a8ad86; engine

receipt: sha256:80e44dc1ceefce9f9a613530d8d23f3772b0bb3748587e4035c9302b56b8d294. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

46. Collective cascade

Claim: SFT-SOCIAL-CASCADE-001 **Family:** network_diffusion_transition **Exact statement:** A cascade retains ordered local transitions, paths, nontransitions and external causes.

Forward chain and exact carrier

The generated carrier is `agents-state-threshold-path-carrier`. Its required relation is `local-transition-propagation-relation`. The reconstruction must retain `order-links-opportunities-nontransitions-and-boundaries-held`. Its declared empirical boundary is `cascade-event-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GATEKEEPING-001, SFT-SOCIAL-SYNCHRONIZATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-state-threshold-path-carrier`.
- **boundary** - which population, place, period and method bound the statement: `cascade-event-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `local-transition-propagation-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `order-links-opportunities-nontransitions-and-boundaries-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`agents-state-threshold-path-carrier__cascade-event-bounded__local-transition-propagation-relation__order-links-opportunities-nontransitions-and-boundaries-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject aggregate change without an observed transmission chain as a cascade.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256: 5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective cascade**, the consequence is practical and exact: A cascade retains ordered local transitions, paths, nontransitions and external causes. A scientific claim therefore has to expose the carrier `agents-state-threshold-path-carrier`, the relation `local-transition-propagation-relation`, the record `order-links-opportunities-nontransitions-and-boundaries-held`, and the boundary `cascade-event-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 4f03311bc088ecf179f3cdd1999442e4fd5643fce0c3789a78213a62f9b00ed1; independent certificate: sha256: 703a8bec113535408ad2ab82d4a78bc2b5da0ea97fd4c1869fe5e8d5ab2c7696; engine receipt: sha256: 217adf4d391053fa69d18f11faa90e665f18c827863e7eb1bb41ad447b770c03. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

47. Collective transition

Claim: SFT-SOCIAL-COLLECTIVE-TRANSITION-001 **Family:** network_diffusion_transition **Exact statement:** Collective transition is a population-distribution change with drivers, controls, reversals and historical boundary retained.

Forward chain and exact carrier

The generated carrier is `population-prior-event-later-carrier`. Its required relation is `distribution-change-and-recurrence-relation`. The reconstruction must retain `frame-states-time-drivers-controls-and-reversals-held`. Its declared empirical boundary is `transition-population-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GATEKEEPING-001, SFT-SOCIAL-CASCADE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `population-prior-event-later-carrier`.
- **boundary** - which population, place, period and method bound the statement: `transition-population-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `distribution-change-and-recurrence-relation`.

- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
frame-states-time-drivers-controls-and-reversals-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

population-prior-event-later-carrier__transition-population-bounded__distribution-change-and-recurrence-relation__frame-states-time-drivers-controls-and-reversals-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject one threshold or Fold coordinate as a universal physical social tipping value.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `GSS-NORC-001` - transport captured; captured bytes 67309; snapshot
sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- `ICPSR-DATA-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- `UCDP-DOWNLOADS-001` - transport captured; captured bytes 138078; snapshot
sha256: 5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not

claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective transition**, the consequence is practical and exact: Collective transition is a population-distribution change with drivers, controls, reversals and historical boundary retained. A scientific claim therefore has to expose the carrier **population-prior-event-later-carrier**, the relation **distribution-change-and-recurrence-relation**, the record **frame-states-time-drivers-controls-and-reversals-held**, and the boundary **transition-population-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256:5344fb8c677fa6e2ec536a88efb972600c2c862bd1e971d88a4b7cddd46e1a58; independent
certificate: sha256:486fb98304fe2b574df95dcebbab89d10c615c01cf6ef007758633897b2dee2e; engine
receipt: sha256:2d9c7a6452a51c3691d4ff3f6db6294a483b0723fd357c8e21430ae1a474f510. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

48. Socio-economic cycle

Claim: SFT-SOCIAL-SOCIAL-CYCLE-001 **Family:** network_diffusion_transition **Exact statement:** A social cycle requires recurrence under a declared state definition and retains shocks, broken cycles and measurement change.

Forward chain and exact carrier

The generated carrier is **time-indexed-collective-state-carrier**. Its required relation is **recurrence-and-feedback-relation**. The reconstruction must retain **variables-frame-phases-shocks-breaks-and-time-held**. Its declared empirical boundary is **cycle-series-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GATEKEEPING-001, SFT-SOCIAL-COLLECTIVE-TRANSITION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `time-indexed-collective-state-carrier`.
- **boundary** - which population, place, period and method bound the statement: `cycle-series-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `recurrence-and-feedback-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `variables-frame-phases-shocks-breaks-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
time-indexed-collective-state-carrier__cycle-series-bounded__recurrence-and-feedback-re
lation__variables-frame-phases-shocks-breaks-and-time-held__evidence-and-normative-clas
s-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts_
_no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject two peaks or a narrative phase label as a universal periodic law.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation:

false.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **socio-economic cycle**, the consequence is practical and exact: A social cycle requires recurrence under a declared state definition and retains shocks, broken cycles and measurement change. A scientific claim therefore has to expose the carrier **time-indexed-collective-state-carrier**, the relation **recurrence-and-feedback-relation**, the record **variables-frame-phases-shocks-breaks-and-time-held**, and the boundary **cycle-series-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:da80c3e0458dd776425d3b4740e8bcde870bf4e71917038140cf38dfec9a9d24; independent certificate: sha256:e7982b5485819f0505afe625df8df6a7557cb08a1aba4940183fce13aadaa815; engine receipt: sha256:9959bc1bdebd09230f2219c1b2122637585365e4eca719e1e11ff44ecfbba248. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

49. Culture

Claim: SFT-SOCIAL-CULTURE-001 **Family:** culture_language_transmission **Exact statement:** Culture is historically transmitted and varied practice and meaning, not an immutable property of a labelled group.

Forward chain and exact carrier

The generated carrier is **practices-symbols-groups-history-carrier**. Its required relation is **transmission-variation-and-reproduction-relation**. The reconstruction must retain **members-practices-meanings-variation-sources-and-time-held**. Its declared empirical boundary is **cultural-context-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001,

SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-SOCIAL-CYCLE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `practices-symbols-groups-history-carrier`.
- **boundary** - which population, place, period and method bound the statement: `cultural-context-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `transmission-variation-and-reproduction-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `members-practices-meanings-variation-sources-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`practices-symbols-groups-history-carrier__cultural-context-bounded__transmission-variation-and-reproduction-relation__members-practices-meanings-variation-sources-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject stereotype, majority report or one observer's taxonomy as complete culture.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor

- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **UNESCO-UIS-001** - transport captured; captured bytes 112727; snapshot
sha256: 030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d.
- **GSS-NORC-001** - transport captured; captured bytes 67309; snapshot
sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- **IPUMS-INTERNATIONAL-001** - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **culture**, the consequence is practical and exact: Culture is historically transmitted and varied practice and meaning, not an immutable property of a labelled group. A scientific claim therefore has to expose the carrier `practices-symbols-groups-history-carrier`, the relation `transmission-variation-and-reproduction-relation`, the record `members-practices-meanings-variation-sources-and-time-held`, and the boundary `cultural-context-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256: 00eec0ddc4edfac7b41c078511ada7177ca3504c557ed711c0f0fbd83c271dae; independent
certificate: sha256: 7331bb3d2844eb8686d1aacaf8647bc4baf9c16b6081f273f5ef8ffffee1cc7e; engine
receipt: sha256: e80d173192171fcb88fd26c3c57e7a3191d5ed3d4ca98db62ff54531705b593. The
independent implementation recomputed the candidate product: **True**. All external rows were preserved: **True**.

50. Social symbol and meaning

Claim: `SFT-SOCIAL-SYMBOL-001` **Family:** `culture_language_transmission` **Exact statement:** A symbol's social meaning is user-, context- and history-bound with alternatives retained.

Forward chain and exact carrier

The generated carrier is `form-users-context-interpretation-carrier`. Its required relation is `form-to-meaning-relation`. The reconstruction must retain `form-users-context-alternatives-history-and-change-held`. Its declared empirical boundary is

interpretive-community-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-SOCIAL-CYCLE-001, SFT-SOCIAL-CULTURE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: **form-users-context-interpretation-carrier**.
- **boundary** - which population, place, period and method bound the statement: **interpretive-community-bounded**.
- **relation** - which generated interaction or comparison connects the carriers: **form-to-meaning-relation**.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: **form-users-context-alternatives-history-and-change-held**.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: **evidence-and-normative-class-explicit**.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: **root-bound-forward-forcing**.
- **generality** - whether finite extension preserves every context rather than universalising one case: **positive-finite-extension-retains-all-contexts**.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: **no-extra-rule**.

Their ordered product gives the unique survivor:

form-users-context-interpretation-carrier__interpretive-community-bounded__form-to-meaning-relation__form-users-context-alternatives-history-and-change-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is **depth_independent**. Minimality passed: **True**. Named-form uniqueness passed: **True**. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject a form as carrying one universal social meaning without a community bridge.

- **false_premise** - passed **True**: reject the generated form lacking the complete physical carrier
- **tampered_source** - passed **True**: reject a changed registered source identity
- **tampered_artifact** - passed **True**: reject a missing, duplicate or additional survivor
- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: **source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence**. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **UNESCO-UIS-001** - transport captured; captured bytes 112727; snapshot sha256:030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d.
- **GSS-NORC-001** - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- **IPUMS-INTERNATIONAL-001** - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social symbol and meaning**, the consequence is practical and exact: A symbol's social meaning is user-, context- and history-bound with alternatives retained. A scientific claim therefore has to expose the carrier **form-users-context-interpretation-carrier**, the relation **form-to-meaning-relation**, the record **form-users-context-alternatives-history-and-change-held**, and the boundary **interpretive-community-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:6ea677625fe5f17f1e4ba936ef4dc0fb359ba42b3a669b60c689db807d349f5e; independent certificate: sha256:878fe2b2d32a0fa191bf64ef45c8a4dd2156622fa35c0daea2b31cec00717a53; engine receipt: sha256:2954e69231b375ec0440a69da946ab51192107b05360efbbafe3eba1debc9f84. The independent implementation recomputed the candidate product: **True**. All external rows were preserved: **True**.

51. Language community

Claim: SFT-SOCIAL-LANGUAGE-COMMUNITY-001 **Family:** culture_language_transmission **Exact statement:** A language community retains speakers, varieties, contexts, power, change and corpus limits.

Forward chain and exact carrier

The generated carrier is `speakers-practices-varieties-interactions-carrier`. Its required relation is `linguistic-transmission-and-coordination-relation`. The reconstruction must retain `speakers-varieties-context-power-change-and-access-held`. Its declared empirical boundary is `community-and-corpus-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-SOCIAL-CYCLE-001, SFT-SOCIAL-SYMBOL-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `speakers-practices-varieties-interactions-carrier`.
- **boundary** - which population, place, period and method bound the statement: `community-and-corpus-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `linguistic-transmission-and-coordination-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `speakers-varieties-context-power-change-and-access-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

speakers-practices-varieties-interactions-carrier__community-and-corpus-bounded__linguistic-transmission-and-coordination-relation__speakers-varieties-context-power-change-and-access-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject standard variety, dictionary or state boundary as the whole language community.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UNESCO-UIS-001 - transport captured; captured bytes 112727; snapshot `sha256:030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d`.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot `sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1`.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **language community**, the consequence is practical and exact: A language community retains speakers, varieties, contexts, power, change and corpus limits. A scientific claim therefore has to expose the carrier `speakers-practices-varieties-interactions-carrier`, the relation `linguistic-transmission-and-coordination-relation`, the record `speakers-varieties-context-power-change-and-access-held`, and the boundary `community-and-corpus-bounded` together. If any one is hidden, the result may remain a description, interpretation, model,

political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:a2449c1a58553ce22bfe461991debe7d50416a2c73b03c1aa2dec6a39eed66d4; independent certificate: sha256:f6ec7cc45e1d644c0eb2c8badce0e6a5ec28a36dd480c65beca1547f6ea41cfe; engine receipt: sha256:27f3aec512e0ec3ae29a7826f51de3f4676d678d60865565dcf1e3ea595a536c. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

52. Education and knowledge transmission

Claim: SFT-SOCIAL-EDUCATION-001 **Family:** culture_language_transmission **Exact statement:** Education evidence retains access, instruction, participation, assessment, attrition and context.

Forward chain and exact carrier

The generated carrier is `learners-teachers-content-institution-carrier`. Its required relation is `instruction-participation-learning-relation`. The reconstruction must retain `access-method-content-assessment-attrition-context-and-time-held`. Its declared empirical boundary is `education-setting-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-SOCIAL-CYCLE-001, SFT-SOCIAL-LANGUAGE-COMMUNITY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `learners-teachers-content-institution-carrier`.
- **boundary** - which population, place, period and method bound the statement: `education-setting-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `instruction-participation-learning-relation`.

- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
access-method-content-assessment-attribution-context-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

learners-teachers-content-institution-carrier__education-setting-bounded__instruction-participation-learning-relation__access-method-content-assessment-attribution-context-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is **depth_independent**. Minimality passed: **True**. Named-form uniqueness passed: **True**. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject credential attainment as complete learning or nonattendance as inability.

- **false_premise** - passed **True**: reject the generated form lacking the complete physical carrier
- **tampered_source** - passed **True**: reject a changed registered source identity
- **tampered_artifact** - passed **True**: reject a missing, duplicate or additional survivor
- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: **source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence**. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **UNESCO-UIS-001** - transport **captured**; captured bytes 112727; snapshot
sha256: 030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d.
- **GSS-NORC-001** - transport **captured**; captured bytes 67309; snapshot
sha256: f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- **IPUMS-INTERNATIONAL-001** - transport **failed_preserved**; captured bytes 0; snapshot **unresolved** and **preserved**.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **education and knowledge transmission**, the consequence is practical and exact: Education evidence retains access, instruction, participation, assessment, attrition and context. A scientific claim therefore has to expose the carrier learners-teachers-content-institution-carrier, the relation instruction-participation-learning-relation, the record access-method-content-assessment-attrition-context-and-time-held, and the boundary education-setting-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256: fcbdc4ac46e81898a804d663fb56755e5381c3562bc06677cbda0f7d0ee89d25; independent
certificate: sha256: 10184ba80f4dbbd8bdb5ad7134820662ef01ff47543cad6dd6cfbe86f6e3004; engine
receipt: sha256: c3e3422783196463271d69df0f596ae0d30a3b0485b0ba8f0723b8787e235e0d. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

53. Media system

Claim: SFT-SOCIAL-MEDIA-001 **Family:** culture_language_transmission **Exact statement:** A media system retains ownership, production, selection, distribution, reach, response and missing exposure.

Forward chain and exact carrier

The generated carrier is producers-content-platform-audience-carrier. Its required relation is production-selection-distribution-relation. The reconstruction must retain ownership-algorithms-content-reach-responses-and-missing-held. Its declared empirical boundary is media-system-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-SOCIAL-CYCLE-001, SFT-SOCIAL-EDUCATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `producers-content-platform-audience-carrier`.
- **boundary** - which population, place, period and method bound the statement: `media-system-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `production-selection-distribution-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `ownership-algorithms-content-reach-responses-and-missing-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
producers-content-platform-audience-carrier__media-system-bounded__production-selection-
-distribution-relation__ownership-algorithms-content-reach-responses-and-missing-held__
evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-exte
nsion-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject published content as received information or platform metrics as all social response.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UNESCO-UIS-001 - transport captured; captured bytes 112727; snapshot sha256:030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **media system**, the consequence is practical and exact: A media system retains ownership, production, selection, distribution, reach, response and missing exposure. A scientific claim therefore has to expose the carrier `producers-content-platform-audience-carrier`, the relation `production-selection-distribution-relation`, the record `ownership-algorithms-content-reach-responses-and-missing-held`, and the boundary `media-system-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:705a33e6d16c00e608670dacf06b9ec4bb644df9541bfe64d06b044fb60efb58; independent certificate: sha256:30437a814bb684453f210b41a9118d0a779e9dbfbae909f1206e41834595c563; engine receipt: sha256:d746622ff5a470f4d18e70033d6bbf1a9ca1377736df7efca8193c286527fbe1. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

54. Collective memory

Claim: SFT-SOCIAL-COLLECTIVE-MEMORY-001 **Family:** `culture_language_transmission` **Exact statement:** Collective memory is a contested transmission of records, narratives and practices with omissions retained.

Forward chain and exact carrier

The generated carrier is `groups-records-narratives-practices-carrier`. Its required relation is `selection-transmission-and-contestation-relation`. The reconstruction must retain `sources-custody-omissions-versions-contestation-and-time-held`. Its declared empirical boundary is `memory-community-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,

SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001,
 SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001,
 SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001,
 SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001,
 SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001,
 SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001,
 SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001,
 SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001,
 SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001,
 SFT-SOCIAL-SOCIAL-CYCLE-001, SFT-SOCIAL-MEDIA-001. Each dependency was model-admitted before this claim.
 The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `groups-records-narratives-practices-carrier`.
- **boundary** - which population, place, period and method bound the statement: `memory-community-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `selection-transmission-and-contestation-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `sources-custody-omissions-versions-contestation-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`groups-records-narratives-practices-carrier__memory-community-bounded__selection-transmission-and-contestation-relation__sources-custody-omissions-versions-contestation-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject dominant narrative as complete history or a destroyed record as evidence of no event.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UNESCO-UIS-001 - transport captured; captured bytes 112727; snapshot
sha256:030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective memory**, the consequence is practical and exact: Collective memory is a contested transmission of records, narratives and practices with omissions retained. A scientific claim therefore has to expose the carrier groups-records-narratives-practices-carrier, the relation selection-transmission-and-contestation-relation, the record sources-custody-omissions-versions-contestation-and-time-held, and the boundary memory-community-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:81f760e248518bd7808ff4361d4be0a4a895477a1b6d168860b8c683c62b0142; independent certificate: sha256:e80d827e7845e127ba3bccd64d8ba48c5ce0b3c4e3ce173f5ffe6213a410c36a; engine receipt: sha256:787f1074b105c114d5e3506c01c9a626a39f67ac8a73f37ee6f75893632723e2. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

55. Individual-to-collective composition

Claim: SFT-SOCIAL-MICRO-MACRO-001 **Family:** multi_level_causation **Exact statement:** Individual states compose a collective only through an explicit relation and aggregation rule retaining lost distinctions.

Forward chain and exact carrier

The generated carrier is `agents-relations-aggregate-carrier`. Its required relation is `composition-and-distribution-relation`. The reconstruction must retain `agent-states-links-weights-aggregation-loss-and-time-held`. Its declared empirical boundary is `composition-rule-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-MEMORY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-relations-aggregate-carrier`.
- **boundary** - which population, place, period and method bound the statement: `composition-rule-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `composition-and-distribution-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `agent-states-links-weights-aggregation-loss-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`agents-relations-aggregate-carrier__composition-rule-bounded__composition-and-distribution-relation__agent-states-links-weights-aggregation-loss-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-`

all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is **depth_independent**. Minimality passed: **True**. Named-form uniqueness passed: **True**. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject an average as a causal collective agent or infer each individual from the aggregate.

- **false_premise** - passed **True**: reject the generated form lacking the complete physical carrier
- **tampered_source** - passed **True**: reject a changed registered source identity
- **tampered_artifact** - passed **True**: reject a missing, duplicate or additional survivor
- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: **source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence**. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **IPUMS-INTERNATIONAL-001** - transport **failed_preserved**; captured bytes 0; snapshot **unresolved** and **preserved**.
- **WORLD-BANK-WDI-001** - transport **captured**; captured bytes 1874098; snapshot **sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927**.
- **WHO-GHO-001** - transport **captured**; captured bytes 141734; snapshot **sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f**.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **individual-to-collective composition**, the consequence is practical and exact: Individual states compose a collective only through an explicit relation and aggregation rule retaining lost distinctions. A scientific claim therefore has to expose the carrier **agents-relations-aggregate-carrier**, the relation **composition-and-distribution-relation**, the record **agent-states-links-weights-aggregation-loss-and-time-held**, and the boundary **composition-rule-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must

still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:2994882565a7ebc4b98a68483553bf9bb856ba11d7edfb7f458a4b15cc64dabb; independent certificate: sha256:0c68bd341fb81524da139fcf7628e5ccd170394da760e639cf88843ceda5d24a; engine receipt: sha256:cff9adcd8d1aa01d9fe52c2fe7b60bd3275ac72a813de37b7ffcc9cf2ae99eb0. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

56. Collective-to-individual constraint

Claim: SFT-SOCIAL-MACRO-MICRO-001 **Family:** multi_level_causation **Exact statement:** Collective structures affect individuals through observable rules, resources, exposure and expectations.

Forward chain and exact carrier

The generated carrier is `institution-agent-options-action-carrier`. Its required relation is `rule-resource-and-expectation-constraint-relation`. The reconstruction must retain `institution-options-exposure-capability-response-and-time-held`. Its declared empirical boundary is `constraint-context-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-MEMORY-001, SFT-SOCIAL-MICRO-MACRO-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `institution-agent-options-action-carrier`.
- **boundary** - which population, place, period and method bound the statement: `constraint-context-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `rule-resource-and-expectation-constraint-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `institution-options-exposure-capability-response-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.

- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

institution-agent-options-action-carrier__constraint-context-bounded__rule-resource-and-
-expectation-constraint-relation__institution-options-exposure-capability-response-and-
time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-
finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is **depth_independent**. Minimality passed: **True**. Named-form uniqueness passed: **True**. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject a macro label as direct cause without a transmission path.

- **false_premise** - passed **True**: reject the generated form lacking the complete physical carrier
- **tampered_source** - passed **True**: reject a changed registered source identity
- **tampered_artifact** - passed **True**: reject a missing, duplicate or additional survivor
- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: **source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence**. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **IPUMS-INTERNATIONAL-001** - transport **failed_preserved**; captured bytes 0; snapshot **unresolved** and **preserved**.
- **WORLD-BANK-WDI-001** - transport **captured**; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- **WHO-GHO-001** - transport **captured**; captured bytes 141734; snapshot
sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective-to-individual constraint**, the consequence is practical and exact: Collective structures affect individuals through observable rules, resources, exposure and expectations. A scientific claim therefore has to expose the carrier **institution-agent-options-action-carrier**, the relation **rule-resource-and-expectation-constraint-relation**, the record **institution-options-exposure-capability-response-and-time-held**, and the boundary **constraint-context-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:b3950fa9b3e081ec0c83406b18c7b85c3eead79f0b85cabfc53ee0820d0a576b; independent certificate: sha256:d5c2e489201f063085428ae65370cf714079355f0bd5d5bad2929120d359609b; engine receipt: sha256:f67cabe097f5789ea4637b873b80b854246c72619c73ef360a3cf62538e7f06b. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

57. Collective emergence

Claim: SFT-SOCIAL-EMERGENCE-001 **Family:** multi_level_causation **Exact statement:** An emergent collective property is composition-dependent and not identical to any one component, with tests and scale retained.

Forward chain and exact carrier

The generated carrier is **agents-relations-new-collective-property-carrier**. Its required relation is **composition-and-nonidentity-relation**. The reconstruction must retain **components-links-property-tests-alternatives-and-scale-held**. Its declared empirical boundary is **emergence-definition-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-MEMORY-001, SFT-SOCIAL-MACRO-MICRO-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-relations-new-collective-property-carrier`.
- **boundary** - which population, place, period and method bound the statement: `emergence-definition-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `composition-and-nonidentity-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `components-links-property-tests-alternatives-and-scale-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
agents-relations-new-collective-property-carrier__emergence-definition-bounded__composition-and-nonidentity-relation__components-links-property-tests-alternatives-and-scale-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject complexity language as proof or erase component contributions.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective emergence**, the consequence is practical and exact: An emergent collective property is composition-dependent and not identical to any one component, with tests and scale retained. A scientific claim therefore has to expose the carrier agents-relations-new-collective-property-carrier, the relation composition-and-nonidentity-relation, the record components-links-property-tests-alternatives-and-scale-held, and the boundary emergence-definition-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:3f76a9f8ee810f734a9f1478a5c24b937eb59fd319e28dd60d32679fd229f995; independent certificate: sha256:2efacbf799260066515864ba843bfe383153f5b52affc4b72d9f603a5fe90aae; engine receipt: sha256:539df59af1716049ceaeec48ce23f8df321d8041e8ce99f262e5517c81ea252c. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

58. Human and artificial multi-agent collective

Claim: SFT-SOCIAL-MULTI-AGENT-001 **Family:** multi_level_causation **Exact statement:** A mixed collective retains human and artificial agent types, protocols, resources, failures and oversight.

Forward chain and exact carrier

The generated carrier is heterogeneous-agents-protocol-environment-carrier. Its required relation is interaction-adaptation-and-outcome-relation. The reconstruction must retain agent-types-rules-resources-actions-failures-and-oversight-held. Its declared empirical boundary is system-and-deployment-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001,

SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-MEMORY-001, SFT-SOCIAL-EMERGENCE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: heterogeneous-agents-protocol-environment-carrier.
- **boundary** - which population, place, period and method bound the statement: system-and-deployment-bounded.
- **relation** - which generated interaction or comparison connects the carriers: interaction-adaptation-and-outcome-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: agent-types-rules-resources-actions-failures-and-oversight-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

heterogeneous-agents-protocol-environment-carrier__system-and-deployment-bounded__interaction-adaptation-and-outcome-relation__agent-types-rules-resources-actions-failures-and-oversight-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject simulated coordination as real-world governance evidence or opaque agents as lawful social causes.

- **false_premise** - passed `True`: reject the generated form lacking the complete physical carrier
- **tampered_source** - passed `True`: reject a changed registered source identity
- **tampered_artifact** - passed `True`: reject a missing, duplicate or additional survivor

- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **IPUMS-INTERNATIONAL-001** - transport **failed_preserved**; captured bytes 0; snapshot **unresolved** and **preserved**.
- **WORLD-BANK-WDI-001** - transport **captured**; captured bytes 1874098; snapshot `sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927`.
- **WHO-GHO-001** - transport **captured**; captured bytes 141734; snapshot `sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **human and artificial multi-agent collective**, the consequence is practical and exact: A mixed collective retains human and artificial agent types, protocols, resources, failures and oversight. A scientific claim therefore has to expose the carrier `heterogeneous-agents-protocol-environment-carrier`, the relation `interaction-adaptation-and-outcome-relation`, the record `agent-types-rules-resources-actions-failures-and-oversight-held`, and the boundary `system-and-deployment-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: `sha256:e487a5f0d5cc1e4532cc082019dc1670104767e15ea65951ca73c995830afac0`; independent certificate: `sha256:362a4f3bad4b21c8118a4c320bcf1ab9659d9b017152f6a1c1ba6204fb5ff239`; engine receipt: `sha256:60e78a62c2a13e7aca9740c5c8c924fddaf5177f7b747adcfc487785ccf4067b`. The independent implementation recomputed the candidate product: **True**. All external rows were preserved: **True**.

59. Collective decision

Claim: **SFT-SOCIAL-COLLECTIVE-DECISION-001** **Family:** `multi_level_causation` **Exact statement:** Collective decision retains eligibility, options, information, rule, participation, dissent and implementation.

Forward chain and exact carrier

The generated carrier is `participants-options-rule-outcome-carrier`. Its required relation is `preference-input-to-decision-relation`. The reconstruction must retain `eligibility-options-information-rule-participation-dissent-and-time-held`. Its declared empirical

boundary is **decision-procedure-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-MEMORY-001, SFT-SOCIAL-MULTI-AGENT-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: **participants-options-rule-outcome-carrier**.
- **boundary** - which population, place, period and method bound the statement: **decision-procedure-bounded**.
- **relation** - which generated interaction or comparison connects the carriers: **preference-input-to-decision-relation**.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: **eligibility-options-information-rule-participation-dissent-and-time-held**.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: **evidence-and-normative-class-explicit**.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: **root-bound-forward-forcing**.
- **generality** - whether finite extension preserves every context rather than universalising one case: **positive-finite-extension-retains-all-contexts**.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: **no-extra-rule**.

Their ordered product gives the unique survivor:

participants-options-rule-outcome-carrier__decision-procedure-bounded__preference-input-to-decision-relation__eligibility-options-information-rule-participation-dissent-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and

the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is **depth_independent**. Minimality passed: **True**. Named-form uniqueness passed: **True**. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject majority outcome as unanimous preference, truth or ethical legitimacy.

- **false_premise** - passed **True**: reject the generated form lacking the complete physical carrier
- **tampered_source** - passed **True**: reject a changed registered source identity
- **tampered_artifact** - passed **True**: reject a missing, duplicate or additional survivor
- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: **source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence**. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **IPUMS-INTERNATIONAL-001** - transport **failed_preserved**; captured bytes 0; snapshot **unresolved** and **preserved**.
- **WORLD-BANK-WDI-001** - transport **captured**; captured bytes 1874098; snapshot **sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927**.
- **WHO-GHO-001** - transport **captured**; captured bytes 141734; snapshot **sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f**.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective decision**, the consequence is practical and exact: Collective decision retains eligibility, options, information, rule, participation, dissent and implementation. A scientific claim therefore has to expose the carrier **participants-options-rule-outcome-carrier**, the relation **preference-input-to-decision-relation**, the record **eligibility-options-information-rule-participation-dissent-and-time-held**, and the boundary **decision-procedure-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:29f156890e5dd163e5fa6cdf8660805384bc59c0d4eb6a8721002c26320a7250; independent

certificate: **sha256:b135dac0f958b4c168b6806489ed99046cb56688c144e7d5ca2e803b6cf62ef1**; engine

receipt: sha256:ebd6ba8b976fd3197bf996059944d8bcbead597c2e03840e9f07c0a3e356861b. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

60. Aggregation and ecological inference boundary

Claim: SFT-SOCIAL-AGGREGATION-LOSS-001 **Family:** multi_level_causation **Exact statement:** Aggregation explicitly records the many-to-one loss preventing recovery of individual states from totals.

Forward chain and exact carrier

The generated carrier is `individual-records-aggregate-carrier`. Its required relation is `many-to-one-summary-relation`. The reconstruction must retain `source-rows-rule-distribution-loss-and-uncertainty-held`. Its declared empirical boundary is `aggregation-level-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-MEMORY-001, SFT-SOCIAL-COLLECTIVE-DECISION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `individual-records-aggregate-carrier`.
- **boundary** - which population, place, period and method bound the statement: `aggregation-level-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `many-to-one-summary-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `source-rows-rule-distribution-loss-and-uncertainty-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

individual-records-aggregate-carrier__aggregation-level-bounded__many-to-one-summary-relation__source-rows-rule-distribution-loss-and-uncertainty-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject ecological or atomistic inference without a cross-level bridge.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `IPUMS-INTERNATIONAL-001` - transport `failed_preserved`; captured bytes 0; snapshot unresolved and preserved.
- `WORLD-BANK-WDI-001` - transport `captured`; captured bytes 1874098; snapshot `sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927`.
- `WHO-GHO-001` - transport `captured`; captured bytes 141734; snapshot `sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **aggregation and ecological inference boundary**, the consequence is practical and exact: Aggregation explicitly records the many-to-one loss preventing recovery of individual states from totals. A scientific claim therefore has to expose the carrier `individual-records-aggregate-carrier`, the relation `many-to-one-summary-relation`, the record `source-rows-rule-distribution-loss-and-uncertainty-held`, and the boundary `aggregation-level-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:b4bd4bd7c53a3187caef27b8737cc88c11ad9b1cc5ca3671fdd645046f88af8e; independent certificate: sha256:68db722be074104ce183ced13bcfcb9888a3d562556b9546cdcaa9fa9d631718; engine receipt: sha256:7ff9a069d479701ef5f32b29468908f4476ca0bcd77572f62d413cdd13c32765. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

61. Historical social event

Claim: SFT-SOCIAL-HISTORICAL-EVENT-001 **Family:** history_intervention_ethics **Exact statement:** A historical event is a source-custodied reconstruction retaining perspectives, gaps and competing accounts.

Forward chain and exact carrier

The generated carrier is `actors-actions-records-context-carrier`. Its required relation is `source-to-event-reconstruction`. The reconstruction must retain `sources-custody-dates-perspectives-gaps-and-alternatives-held`. Its declared empirical boundary is `event-and-source-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-AGGREGATION-LOSS-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `actors-actions-records-context-carrier`.
- **boundary** - which population, place, period and method bound the statement: `event-and-source-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `source-to-event-reconstruction`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `sources-custody-dates-perspectives-gaps-and-alternatives-held`.

- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

actors-actions-records-context-carrier__event-and-source-bounded__source-to-event-reconstruction__sources-custody-dates-perspectives-gaps-and-alternatives-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject a single archive, victor narrative or later outcome as the complete event.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `UCDP-DOWNLOADS-001` - transport captured; captured bytes 138078; snapshot sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- `VDEM-DATA-001` - transport captured; captured bytes 122924; snapshot sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- `ICPSR-DATA-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and

adverse rows retained.

Scientific meaning

For **historical social event**, the consequence is practical and exact: A historical event is a source-custodied reconstruction retaining perspectives, gaps and competing accounts. A scientific claim therefore has to expose the carrier **actors-actions-records-context-carrier**, the relation **source-to-event-reconstruction**, the record **sources-custody-dates-perspectives-gaps-and-alternatives-held**, and the boundary **event-and-source-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:249a0e97a0fdd4a27fe064ef274fb018703e8e4b79931a92c66729ff4ea4684a; independent certificate: sha256:3f26c28018ad05367627d629669f148dac685b51f74d642ff4f1058f07adc778; engine receipt: sha256:5d2c8091d96cfda76c054ea5ab140b4534508d6d7f8d2c64d102934786854533. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

62. Historical contingency

Claim: SFT-SOCIAL-CONTINGENCY-001 **Family:** history_intervention_ethics **Exact statement:** Contingency retains prior alternatives, event sequence, mechanism and unresolved counterfactuals.

Forward chain and exact carrier

The generated carrier is **prior-state-event-alternative-later-carrier**. Its required relation is **path-dependent-transition-relation**. The reconstruction must retain **prior-options-events-sequence-mechanisms-and-unknowns-held**. Its declared empirical boundary is **case-and-period-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-AGGREGATION-LOSS-001, SFT-SOCIAL-HISTORICAL-EVENT-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `prior-state-event-alternative-later-carrier`.
- **boundary** - which population, place, period and method bound the statement: `case-and-period-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `path-dependent-transition-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `prior-options-events-sequence-mechanisms-and-unknowns-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`prior-state-event-alternative-later-carrier__case-and-period-bounded__path-dependent-transition-relation__prior-options-events-sequence-mechanisms-and-unknowns-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject inevitability from hindsight or universal law from one historical path.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **historical contingency**, the consequence is practical and exact: Contingency retains prior alternatives, event sequence, mechanism and unresolved counterfactuals. A scientific claim therefore has to expose the carrier prior-state-event-alternative-later-carrier, the relation path-dependent-transition-relation, the record prior-options-events-sequence-mechanisms-and-unknowns-held, and the boundary case-and-period-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:71ea865ffc8a79e0416e91d196ee64f318638ed84e66467153971cf70585c9e5; independent certificate: sha256:88fa6aecf2f8056955cd03d7d0aa76219c3fceff6700bd0a319d23225b32119b; engine receipt: sha256:79be04ee3a298f8ceca4cf7827cb30d905f125227f92f742158487be5c98a0c1. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

63. Social causal inference

Claim: SFT-SOCIAL-CAUSAL-INFERENC-001 **Family:** history_intervention_ethics **Exact statement:** Causal inference retains assignment, comparison, confounding, spillover, attrition and target population.

Forward chain and exact carrier

The generated carrier is units-exposure-comparison-outcome-carrier. Its required relation is counterfactual-comparison-relation. The reconstruction must retain assignment-selection-confounding-time-attrition-and-spillover-held. Its declared empirical boundary is design-and-population-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001,

SFT-SOCIAL-AGGREGATION-LOSS-001, SFT-SOCIAL-CONTINGENCY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `units-exposure-comparison-outcome-carrier`.
- **boundary** - which population, place, period and method bound the statement: `design-and-population-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `counterfactual-comparison-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `assignment-selection-confounding-time-attrition-and-spillover-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`units-exposure-comparison-outcome-carrier__design-and-population-bounded__counterfactual-comparison-relation__assignment-selection-confounding-time-attrition-and-spillover-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject association, model coefficient or adjusted significance as causation without design support.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot sha256: 5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot sha256: cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social causal inference**, the consequence is practical and exact: Causal inference retains assignment, comparison, confounding, spillover, attrition and target population. A scientific claim therefore has to expose the carrier `units-exposure-comparison-outcome-carrier`, the relation `counterfactual-comparison-relation`, the record `assignment-selection-confounding-time-attrition-and-spillover-held`, and the boundary `design-and-population-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256: 4cd7bc322a0db065a643484da67d0ad1e4f63f66694df43856ca962b8dfdfb2f; independent certificate: sha256: 0571c271a87d3c13f994c878a4f159bd8ab7da52a268df72b003b222930c0bc6; engine receipt: sha256: 2ed40b2f01083326f219d9aca6f455c1d6585f185b410b751a7e4bcf18bb737e. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

64. Social intervention

Claim: SFT-SOCIAL-INTERVENTION-001 **Family:** `history_intervention_ethics` **Exact statement:** Intervention evidence retains target, delivery, uptake, context, comparison, harms and missing outcomes.

Forward chain and exact carrier

The generated carrier is `population-action-implementation-outcome-carrier`. Its required relation is `implementation-to-outcome-comparison`. The reconstruction must retain `target-delivery-uptake-context-harms-and-missing-held`. Its declared empirical boundary is `intervention-context-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-AGGREGATION-LOSS-001, SFT-SOCIAL-CAUSAL-INFERENCE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `population-action-implementation-outcome-carrier`.
- **boundary** - which population, place, period and method bound the statement: `intervention-context-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `implementation-to-outcome-comparison`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `target-delivery-uptake-context-harms-and-missing-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`population-action-implementation-outcome-carrier__intervention-context-bounded__implementation-to-outcome-comparison__target-delivery-uptake-context-harms-and-missing-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is **depth_independent**. Minimality passed: **True**. Named-form uniqueness passed: **True**. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject policy announcement as implementation or average benefit as absence of harm.

- **false_premise** - passed **True**: reject the generated form lacking the complete physical carrier
- **tampered_source** - passed **True**: reject a changed registered source identity
- **tampered_artifact** - passed **True**: reject a missing, duplicate or additional survivor
- **boundary** - passed **True**: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: **source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence**. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: **True**. External evidence selected the survivor: **false**. Credential or prestige was used as evidence: **false**. Consensus was used as proof: **false**. Normative judgment was relabelled observation: **false**.

- **UCDP-DOWNLOADS-001** - transport captured; captured bytes 138078; snapshot sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- **VDEM-DATA-001** - transport captured; captured bytes 122924; snapshot sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- **ICPSR-DATA-001** - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social intervention**, the consequence is practical and exact: Intervention evidence retains target, delivery, uptake, context, comparison, harms and missing outcomes. A scientific claim therefore has to expose the carrier **population-action-implementation-outcome-carrier**, the relation **implementation-to-outcome-comparison**, the record **target-delivery-uptake-context-harms-and-missing-held**, and the boundary **intervention-context-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: a392f51cd50bee633a0c092aa3877e6c3930f4a9911a3b288c624c094e7fa701; independent certificate: sha256: 975ad9bf0c95b677a40fb54eb1eaa43198bb09847162234eec9efcb9210b71ea; engine receipt: sha256: aa62a0c0d10360df013c9b20d813fac4383cb3aff2e18fc51cd9e785bdbfe315. The independent implementation recomputed the candidate product: **True**. All external rows were preserved: **True**.

65. Cross-context replication

Claim: SFT-SOCIAL-REPLICATION-001 **Family:** history_intervention_ethics **Exact statement:** Replication retains populations, settings, measures, protocol deviations, adverse outcomes and transport limits.

Forward chain and exact carrier

The generated carrier is `claim-contexts-protocol-outcomes-carrier`. Its required relation is `protocol-preserving-comparison-relation`. The reconstruction must retain `populations-settings-measures-deviations-results-and-adverse-held`. Its declared empirical boundary is `transportability-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-AGGREGATION-LOSS-001, SFT-SOCIAL-INTERVENTION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `claim-contexts-protocol-outcomes-carrier`.
- **boundary** - which population, place, period and method bound the statement: `transportability-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `protocol-preserving-comparison-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `populations-settings-measures-deviations-results-and-adverse-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

claim-contexts-protocol-outcomes-carrier__transportability-bounded__protocol-preserving-comparison-relation__populations-settings-measures-deviations-results-and-adverse-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject same label with changed construct as replication or one failure as universal impossibility.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `UCDP-DOWNLOADS-001` - transport captured; captured bytes 138078; snapshot `sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5`.
- `VDEM-DATA-001` - transport captured; captured bytes 122924; snapshot `sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c`.
- `ICPSR-DATA-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **cross-context replication**, the consequence is practical and exact: Replication retains populations, settings, measures, protocol deviations, adverse outcomes and transport limits. A scientific claim therefore has to expose the carrier

`claim-contexts-protocol-outcomes-carrier`, the relation `protocol-preserving-comparison-relation`, the record `populations-settings-measures-deviations-results-and-adverse-held`, and the boundary `transportability-bounded` together. If any one is hidden, the result may remain a description, interpretation, model,

political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:18a70a4476b3ee35b194b7f8ccfdd667c0074be162501095795a9fe3e9ddf70e; independent certificate: sha256:f8ac24f95b26a32d2dbb87825021f55b3b97ae7b0057f45eb2338bc42a9fa227; engine receipt: sha256:dd32b12f5464a8ce1964f7e13f600dcc4394a95554cee8702add49c94872de4a. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

66. Ethical and value boundary

Claim: SFT-SOCIAL-ETHICS-BOUNDARY-001 **Family:** history_intervention_ethics **Exact statement:** Ethical judgment retains affected agents, rights, harms, benefits, disagreement and power and is not relabelled an empirical law.

Forward chain and exact carrier

The generated carrier is `affected-agents-values-actions-consequences-carrier`. Its required relation is `normative-deliberation-and-rights-relation`. The reconstruction must retain `stakeholders-values-rights-harms-benefits-disagreement-and-power-held`. Its declared empirical boundary is `ethical-decision-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-AGGREGATION-LOSS-001, SFT-SOCIAL-REPLICATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `affected-agents-values-actions-consequences-carrier`.
- **boundary** - which population, place, period and method bound the statement: `ethical-decision-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `normative-deliberation-and-rights-relation`.

- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
stakeholders-values-rights-harms-benefits-disagreement-and-power-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

affected-agents-values-actions-consequences-carrier__ethical-decision-bounded__normative-deliberation-and-rights-relation__stakeholders-values-rights-harms-benefits-disagreement-and-power-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject empirical frequency as moral permission or moral commitment as measured causal fact.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `UCDP-DOWNLOADS-001` - transport captured; captured bytes 138078; snapshot `sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5`.
- `VDEM-DATA-001` - transport captured; captured bytes 122924; snapshot `sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c`.
- `ICPSR-DATA-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **ethical and value boundary**, the consequence is practical and exact: Ethical judgment retains affected agents, rights, harms, benefits, disagreement and power and is not relabelled an empirical law. A scientific claim therefore has to expose the carrier **affected-agents-values-actions-consequences-carrier**, the relation **normative-deliberation-and-rights-relation**, the record **stakeholders-values-rights-harms-benefits-disagreement-and-power-held**, and the boundary **ethical-decision-bounded** together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 72308cfc9208ff4581dda71edfb062eacb81d0853d418deff376c5bf05591140; independent certificate: sha256: 0571a291a9639095487b59a34be87ecc6ae2a260f849c24db5e1384771bc278f; engine receipt: sha256: 3cfb0486415cdab2c5ce460390aab1782820d675af4b08e782e175442c5e13b3. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

67. Mathematics and graph handoff

Claim: SFT-SOCIAL-MATH-HANDOFF-001 **Family:** ownership_handoffs **Exact statement:** Social Science supplies measured meaning; Mathematics supplies abstract structure without selecting social identity.

Forward chain and exact carrier

The generated carrier is **social-carrier-abstract-structure-carrier**. Its required relation is **typed-abstraction-handoff**. The reconstruction must retain **social-meaning-measurement-map-loss-and-use-held**. Its declared empirical boundary is **disciplinary-ownership-bounded**. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ETHICS-BOUNDARY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `social-carrier-abstract-structure-carrier`.
- **boundary** - which population, place, period and method bound the statement: `disciplinary-ownership-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `typed-abstraction-handoff`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `social-meaning-measurement-map-loss-and-use-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
social-carrier-abstract-structure-carrier__disciplinary-ownership-bounded__typed-abstraction-handoff__social-meaning-measurement-map-loss-and-use-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject an unlabeled graph as a social network law.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256: 84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256: 32ceaf90269046a7bf55fdebd725d304909afac6e76dda8e1016a6d83f3898ba.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256: 4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **mathematics and graph handoff**, the consequence is practical and exact: Social Science supplies measured meaning; Mathematics supplies abstract structure without selecting social identity. A scientific claim therefore has to expose the carrier `social-carrier-abstract-structure-carrier`, the relation `typed-abstraction-handoff`, the record `social-meaning-measurement-map-loss-and-use-held`, and the boundary `disciplinary-ownership-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256: ce19e3e32dc03520abda5709fab08478bb7bd746922a600f5eef94fc1a3a6862; independent
certificate: sha256: b07d808e6f7948ffe4823a000f88871dfd85b650814ad747b692095a42a67c7d; engine
receipt: sha256: fcd6cffbd20780cca6f6c91ae1fc4f054fc9f6357e3612bc6d6d7a869cd5b599b. The
independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

68. Computation and distributed-system handoff

Claim: SFT-SOCIAL-COMPUTATION-HANDOFF-001 **Family:** `ownership_handoffs` **Exact statement:** Computation may simulate a declared social process but cannot make simulated behavior an observation.

Forward chain and exact carrier

The generated carrier is `social-process-computational-model-carrier`. Its required relation is `semantics-preserving-handoff`. The reconstruction must retain `agents-rules-assumptions-model-errors-and-use-held`. Its declared empirical boundary is `model-purpose-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001,

SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ETHICS-BOUNDARY-001, SFT-SOCIAL-MATH-HANDOFF-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `social-process-computational-model-carrier`.
- **boundary** - which population, place, period and method bound the statement: `model-purpose-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `semantics-preserving-handoff`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `agents-rules-assumptions-model-errors-and-use-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`social-process-computational-model-carrier__model-purpose-bounded__semantics-preserving-handoff__agents-rules-assumptions-model-errors-and-use-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject platform behavior or consensus protocol as a universal human law.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier

- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256:32ceaf90269046a7bf55fdebd725d304909afac6e76dda8e1016a6d83f3898ba.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **computation and distributed-system handoff**, the consequence is practical and exact: Computation may simulate a declared social process but cannot make simulated behavior an observation. A scientific claim therefore has to expose the carrier `social-process-computational-model-carrier`, the relation `semantics-preserving-handoff`, the record `agents-rules-assumptions-model-errors-and-use-held`, and the boundary `model-purpose-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:200da5093ee18b4dc8a953ef6eee664db6ed30f644e64e6cce62b119d2d6b2da; independent certificate: sha256:3d21e8532e7bcd4b0323620a1d943ee8c462e61782ffafd3471bd8830bd367b7; engine receipt: sha256:de0cc7b5d201335a2cfa310fc1bc03d706361bd52e26a6eee479e7e16b30a210. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

69. Biology, consciousness and individual cognition handoff

Claim: SFT-SOCIAL-BIO-CONSC-HANDOFF-001 **Family:** `ownership_handoffs` **Exact statement:** Individual mechanisms and experience remain upstream evidence linked to, but not replaced by, collective relations.

Forward chain and exact carrier

The generated carrier is `individual-organism-mind-social-context-carrier`. Its required relation is `cross-level-evidence-handoff`. The reconstruction must retain `mechanism-experience-behaviour-context-and-limit-held`. Its declared empirical boundary is `level-of-analysis-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ETHICS-BOUNDARY-001`, `SFT-SOCIAL-COMPUTATION-HANDOFF-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `individual-organism-mind-social-context-carrier`.
- **boundary** - which population, place, period and method bound the statement: `level-of-analysis-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `cross-level-evidence-handoff`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `mechanism-experience-behaviour-context-and-limit-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`individual-organism-mind-social-context-carrier__level-of-analysis-bounded__cross-level-evidence-handoff__mechanism-experience-behaviour-context-and-limit-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model

or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject biological reduction of an institution or social attribution of a private quale without evidence.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `WHO-GHO-001` - transport captured; captured bytes 141734; snapshot
sha256: 84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.
- `CROSSREF-RECORDS-001` - transport captured; captured bytes 69898; snapshot
sha256: 32ceaf90269046a7bf55fdebd725d304909afac6e76dda8e1016a6d83f3898ba.
- `WORLD-BANK-WDI-001` - transport captured; captured bytes 1874098; snapshot
sha256: 4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **biology, consciousness and individual cognition handoff**, the consequence is practical and exact: Individual mechanisms and experience remain upstream evidence linked to, but not replaced by, collective relations. A scientific claim therefore has to expose the carrier `individual-organism-mind-social-context-carrier`, the relation `cross-level-evidence-handoff`, the record `mechanism-experience-behaviour-context-and-limit-held`, and the boundary `level-of-analysis-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: b352745a8a219c3f808ce3340b1808bcb4821181e6326f935dd5012cc208542a; independent certificate: sha256: 739b84d64a33d4a526397434e53adf10cb1680e635fc48c971288064521e374d; engine receipt: sha256: 7a26afef3725b313ba83e780dc752a863110a77f0e03c86e9d2e6ae177eaac0f. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

70. Medicine and public-health handoff

Claim: SFT-SOCIAL-MEDICINE-PUBLIC-HEALTH-HANDOFF-001 **Family:** ownership_handoffs **Exact statement:** Medicine owns clinical state; Social Science owns collective access and distribution; public-health composition retains both.

Forward chain and exact carrier

The generated carrier is `health-state-population-social-condition-carrier`. Its required relation is `individual-population-and-social-link`. The reconstruction must retain `clinical-measures-exposure-access-distribution-and-context-held`. Its declared empirical boundary is `health-ownership-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ETHICS-BOUNDARY-001, SFT-SOCIAL-BIO-CONSC-HANDOFF-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `health-state-population-social-condition-carrier`.
- **boundary** - which population, place, period and method bound the statement: `health-ownership-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `individual-population-and-social-link`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `clinical-measures-exposure-access-distribution-and-context-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.

- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

health-state-population-social-condition-carrier__health-ownership-bounded__individual-population-and-social-link__clinical-measures-exposure-access-distribution-and-context-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject population association as individual diagnosis or clinical efficacy as equal access.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256: 84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256: 32ceaf90269046a7bf55fdebd725d304909afac6e76dda8e1016a6d83f3898ba.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256: 4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **medicine and public-health handoff**, the consequence is practical and exact: Medicine owns clinical state; Social Science owns collective access and distribution; public-health composition retains both. A scientific claim therefore has to expose the carrier `health-state-population-social-condition-carrier`, the relation `individual-population-and-social-link`, the record `clinical-measures-exposure-access-distribution-and-context-held`, and the boundary `health-ownership-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:b2c21f28a49ac9c99765eb89ac4d244115474b0e53d80e2117071aed2c2a0383; independent certificate: sha256:69afd5d99d267834a5ac23272274a4d9c612b005b6241aa8bed5a9d8f2dadf7a; engine receipt: sha256:2f5748560880bdbbc3ad54e5dd3457d3df4707505ed574df7d6b319f7cc29595. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

71. Earth and environmental handoff

Claim: SFT-SOCIAL-EARTH-HANDOFF-001 **Family:** ownership_handoffs **Exact statement:** Earth Science owns hazard; Social Science owns exposure, vulnerability, collective response and unequal impact.

Forward chain and exact carrier

The generated carrier is `hazard-population-exposure-impact-carrier`. Its required relation is `environment-to-social-impact-chain`. The reconstruction must retain `hazard-exposure-vulnerability-response-time-and-ownership-held`. Its declared empirical boundary is `hazard-impact-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ETHICS-BOUNDARY-001, SFT-SOCIAL-MEDICINE-PUBLIC-HEALTH-HANDOFF-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `hazard-population-exposure-impact-carrier`.
- **boundary** - which population, place, period and method bound the statement: `hazard-impact-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `environment-to-social-impact-chain`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `hazard-exposure-vulnerability-response-time-and-ownership-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`hazard-population-exposure-impact-carrier__hazard-impact-bounded__environment-to-social-impact-chain__hazard-exposure-vulnerability-response-time-and-ownership-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject physical hazard as identical social harm or policy as an Earth law.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `WHO-GHO-001` - transport captured; captured bytes `141734`; snapshot `sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f`.

- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256:32ceaf90269046a7bf55fdedb725d304909afac6e76dda8e1016a6d83f3898ba.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **earth and environmental handoff**, the consequence is practical and exact: Earth Science owns hazard; Social Science owns exposure, vulnerability, collective response and unequal impact. A scientific claim therefore has to expose the carrier hazard-population-exposure-impact-carrier, the relation environment-to-social-impact-chain, the record hazard-exposure-vulnerability-response-time-and-ownership-held, and the boundary hazard-impact-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:12fac511becdbef67e5d84123fe6d053e9b23ba153f6ea3954daa7c96dc01ad7; independent certificate: sha256:fddb59eb2a127570a7094b446dfc5ec0040231e6347ce305e8eb5380d7ee641b; engine receipt: sha256:949369f3908bb753e62f2b5017113fc3d1b5ab7d7d62ab6c26366ce55a227aa0. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

72. Engineering and policy implementation handoff

Claim: SFT-SOCIAL-ENGINEERING-HANDOFF-001 **Family:** ownership_handoffs **Exact statement:** Engineering may test implementations against social laws but applications cannot select or rewrite those laws.

Forward chain and exact carrier

The generated carrier is social-law-design-deployment-consequence-carrier. Its required relation is law-to-implementation-test-relation. The reconstruction must retain requirements-design-users-context-failures-harms-and-feedback-held. Its declared empirical boundary is deployment-purpose-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ETHICS-BOUNDARY-001, SFT-SOCIAL-EARTH-HANDOFF-001. Each dependency was model-admitted

before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `social-law-design-deployment-consequence-carrier`.
- **boundary** - which population, place, period and method bound the statement: `deployment-purpose-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `law-to-implementation-test-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `requirements-design-users-context-failures-harms-and-feedback-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`social-law-design-deployment-consequence-carrier__deployment-purpose-bounded__law-to-implementation-test-relation__requirements-design-users-context-failures-harms-and-feedback-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favorable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavorable controls

The registered falsification condition is: Reject a successful product metric as scientific admission or a failed implementation as automatic law falsification.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from

the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record / primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256: 84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256: 32ceaf90269046a7bf55fdebd725d304909afac6e76dda8e1016a6d83f3898ba.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256: 4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **engineering and policy implementation handoff**, the consequence is practical and exact: Engineering may test implementations against social laws but applications cannot select or rewrite those laws. A scientific claim therefore has to expose the carrier `social-law-design-deployment-consequence-carrier`, the relation `law-to-implementation-test-relation`, the record `requirements-design-users-context-failures-harms-and-feedback-held`, and the boundary `deployment-purpose-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: bb77400f2642664067344c2a6f2c5e24a581e91925ee2042c03bdb3f7d02da35; independent certificate: sha256: 246a3d6fc5d8f22e35a4eda6cb7151641ee9b197ec890e9faf82022b27775c0e; engine receipt: sha256: bfe9b4bbad15adb6a0d7bac2472d99f035c9ac7810b25c365d92b6f5b74b1f87. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

Historical reconciliation

The inherited V1/V2 census contains 763 entries. Every entry was classified atomically by present branch ownership. Five Social-owned questions-collective recurrence, emergence, synchronization and multi-agent composition-were registered as questions and mapped to present claims. Their former answers were not imported as premises. Graph, distributed-computation, cognition, medicine, Earth and engineering questions remain typed handoffs rather than being relabelled Social laws.

What this foundation does and does not claim

It claims that the presently known foundational forms required to state and test Social and Collective Sciences have been generated, uniquely closed, independently reconstructed and source-bound at their declared boundaries. It does not claim that every historical event is known, every archive is accessible, every society shares one parameter, every ethical dispute is solved by

frequency, or all future discovery is closed. It does not turn institutions into persons, collectives into private conscious subjects, simulations into observations or applications into law selectors.

The branch is complete to Maria Smith's current standard and evidence, and open to lawful extension. A new discovery may add a carrier, relation, source, context, historical record or stronger bridge. It must preserve existing adverse, absent and failed records and pass the same engine. No branch is permanently locked against valid additions.

Full-field continuation roadmap

- complete social network group organization and institution census.
- complete anthropology archaeology sociology family and community studies.
- complete governance law politics economics labour and inequality evidence.
- complete demography migration geography city and settlement evidence.
- complete culture language education religion media and knowledge transmission.
- complete criminology justice conflict peace and social-harm evidence.
- complete multi-agent artificial collective and institutional knowledge studies.
- complete causal intervention replication ethics and cross-context limits.

Every later extension will be decomposed into nonduplicate obligations, sealed before outcomes, admitted claim by claim and incorporated through a versioned paper update. Applications, public policy and engineered platforms may test the laws but may not select them.

Reproduction and present validation state

During branch construction the repository uses proportionate checks: both immutable seals around each admission, one 256-form census and independent reconstruction per claim, and the six Social-focused tests after integration. The heavy repository-wide one-command verification is deliberately reserved until every remaining foundation branch has been integrated. This avoids repeatedly consuming reviewer time while preserving the same protected admission authority. The final run will be timed and its completion date and duration recorded in the public documentation.

This branch passed 72 sequential admissions, 72 independent validators, 18,432 candidate decisions, its integration audit and six focused tests. Neither the engine nor verification authority was modified. Remote publication remains unauthorized at this stage; the paper and evidence release are prepared locally for later explicit approval.

Rights, participation and scientific admission

Anyone may read, reproduce, criticize and extend the work. No credential is required to speak. The Ernos Labs designation requires the shared empirical constitution: visible root derivation, exact grammar, complete enumeration, no target selection, no hidden fit, preserved unfavorable evidence, independent reconstruction, unchanged protected authority, retained authorship and open licensing. Criticism is public participation; admission is a separately evidenced result.

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Machine identities

- Inventory: sha256:8b5f0e4c65aa077ab8dad91c3cb00f466068ee387d4f54e63d3df57d56e1745c
- Prior audit: sha256:38d9c04161cffe5f852094afb223a3affc3fbf6776cc89acc86b472c1a7465dc
- Integration audit: sha256:315798d5cebfe65fb8d0a436eaf8171d3386599bd44896f1cff910905a86699
- Source registry: sha256:5ecee0b682729015412e612c6dd71f599dbcb1c4c6a3276b6ac230beab7037af
- External target census:
sha256:f82d4757217304bc64b8210b5f82bc0d301091c38ce358b83d95d460bcfbdf3c

- Engine: sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
- Verification authority: sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8